

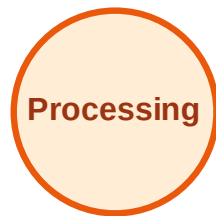
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



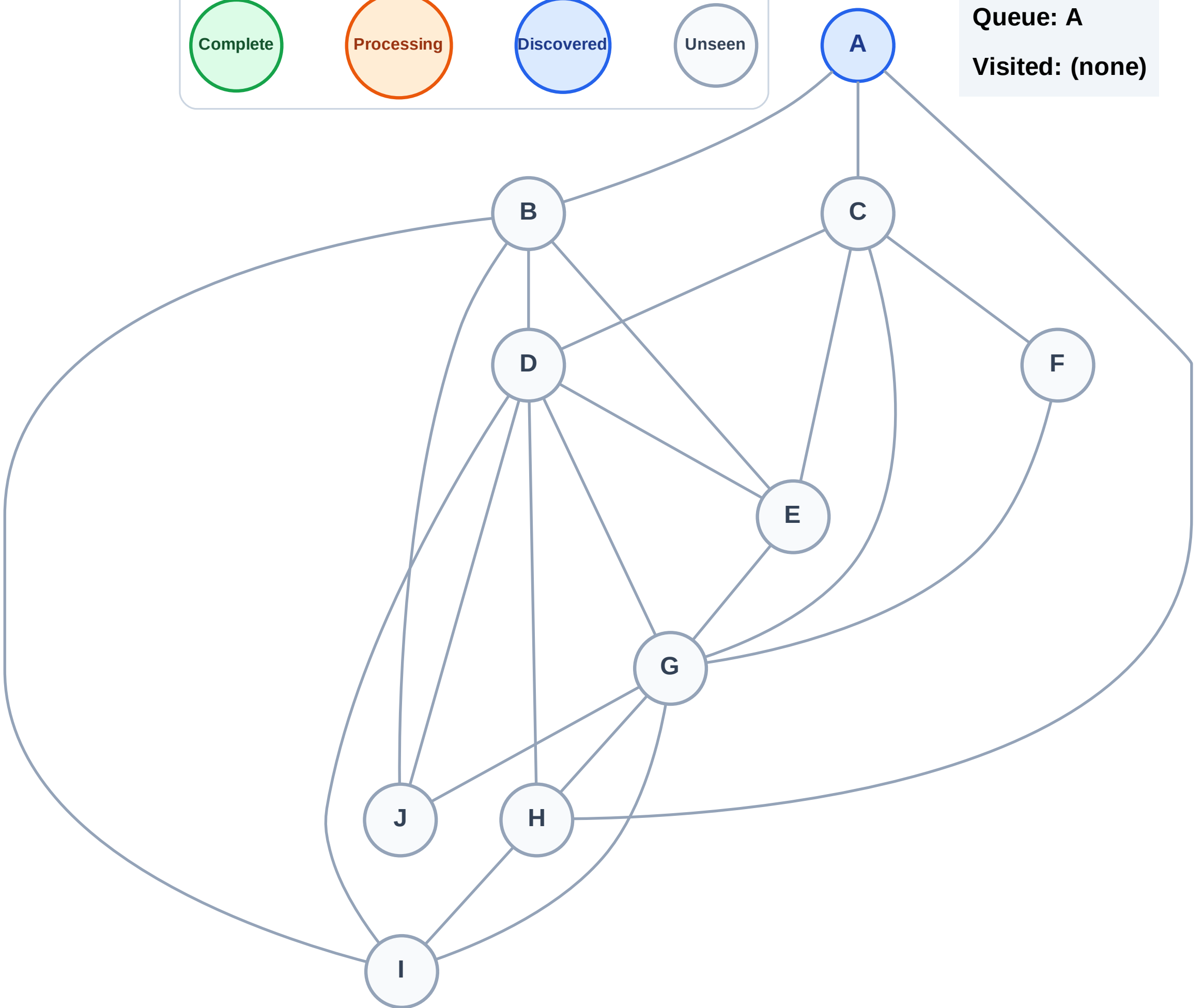
Discovered



Unseen

Queue: A

Visited: (none)



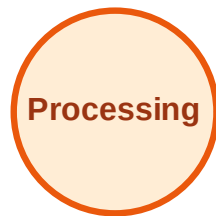
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



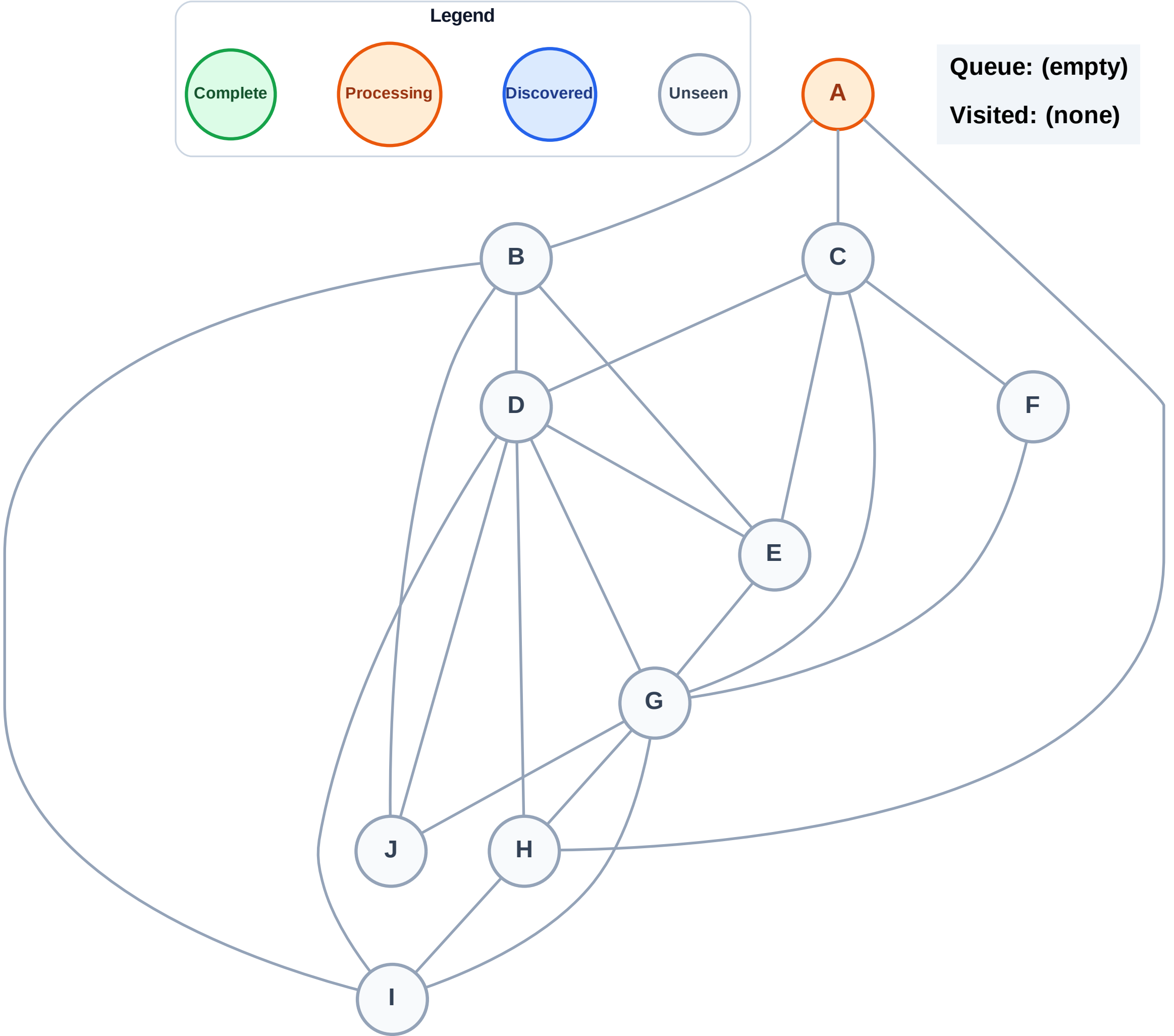
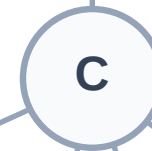
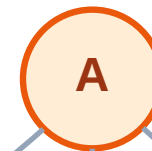
Discovered



Unseen

Queue: (empty)

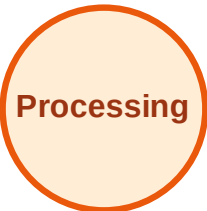
Visited: (none)



BFS Traversal

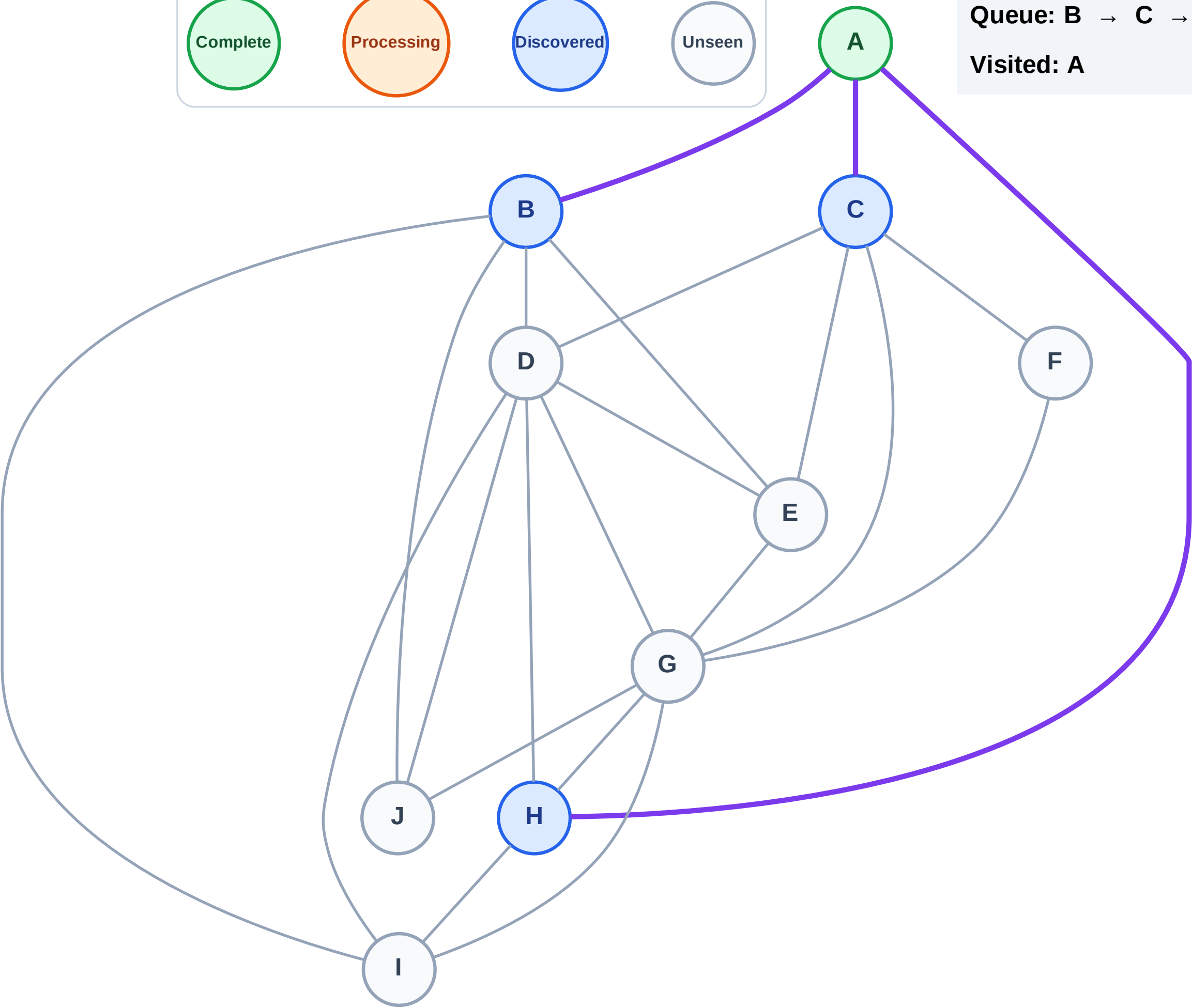
Step 3: Discover B, C, H from A

Legend



Queue: B → C → H

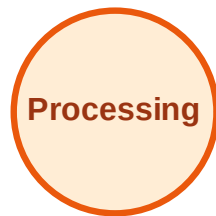
Visited: A



BFS Traversal

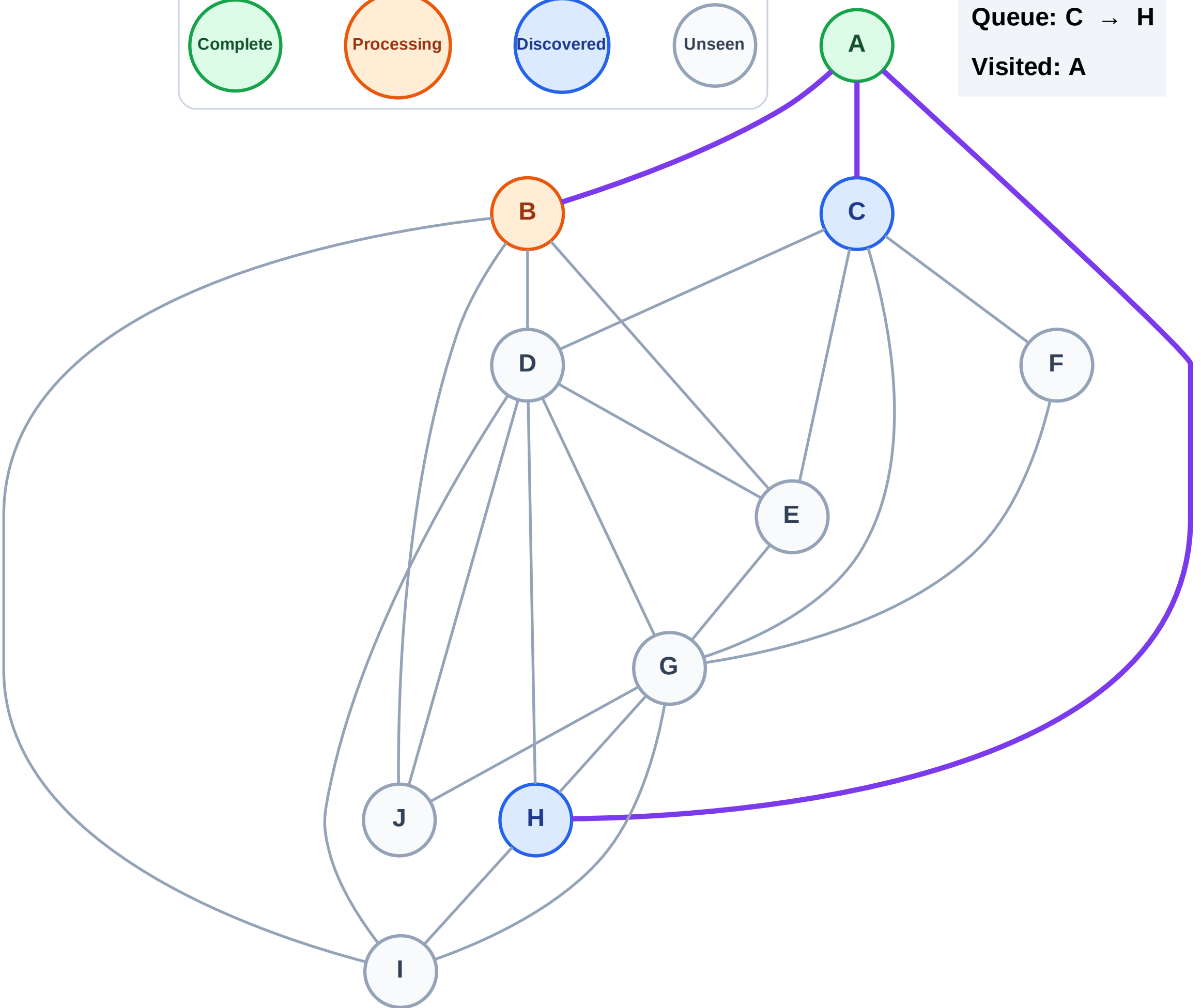
Step 4: Process node B

Legend



Queue: C → H

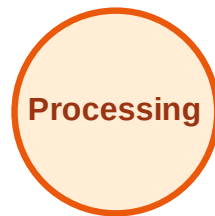
Visited: A



BFS Traversal

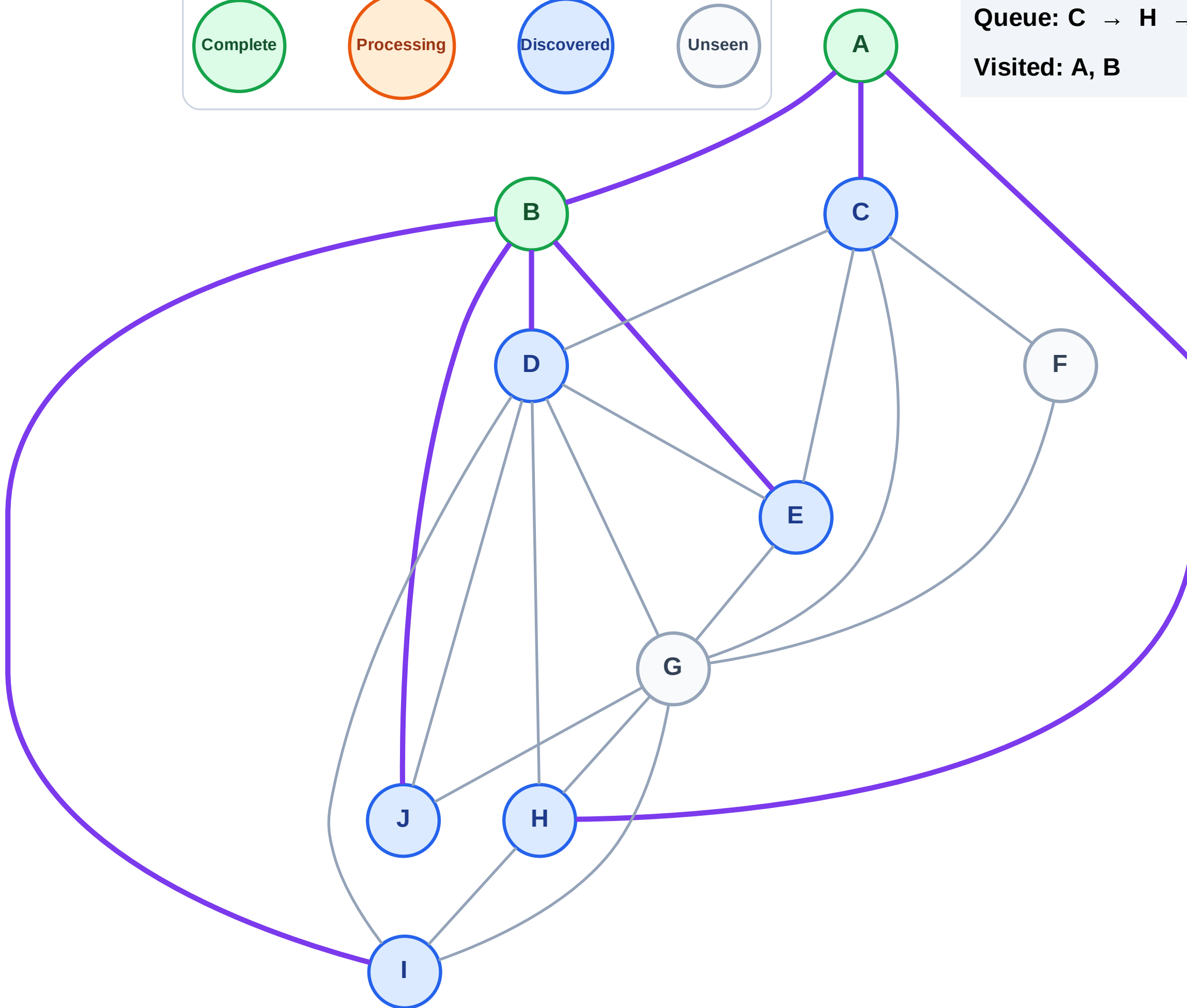
Step 5: Discover D, E, I, J from B

Legend



Queue: C → H → D → E → I → J

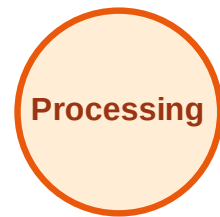
Visited: A, B



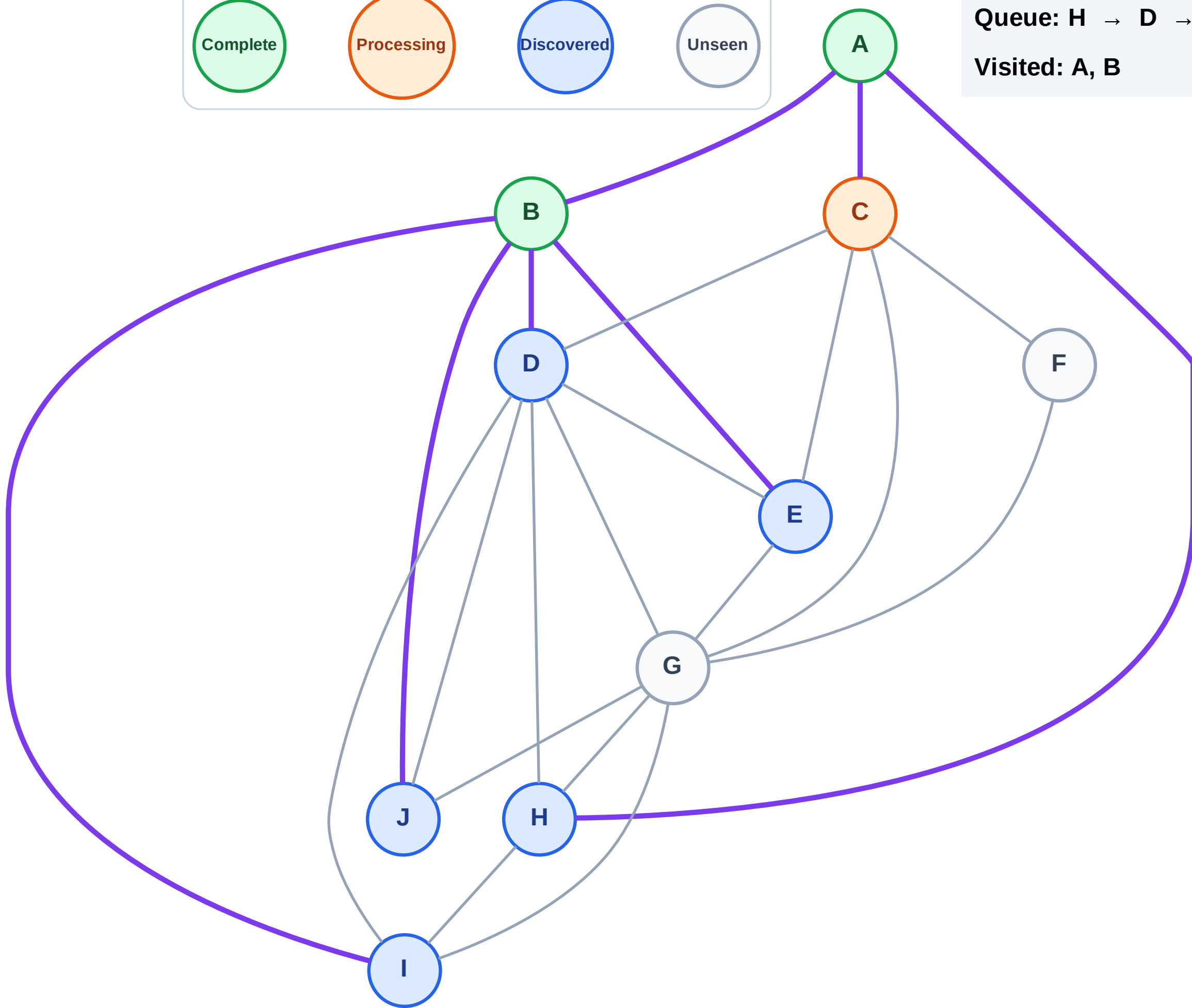
BFS Traversal

Step 6: Process node C

Legend



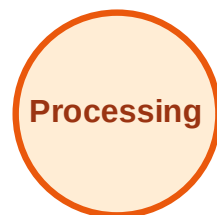
Queue: H → D → E → I → J
Visited: A, B



BFS Traversal

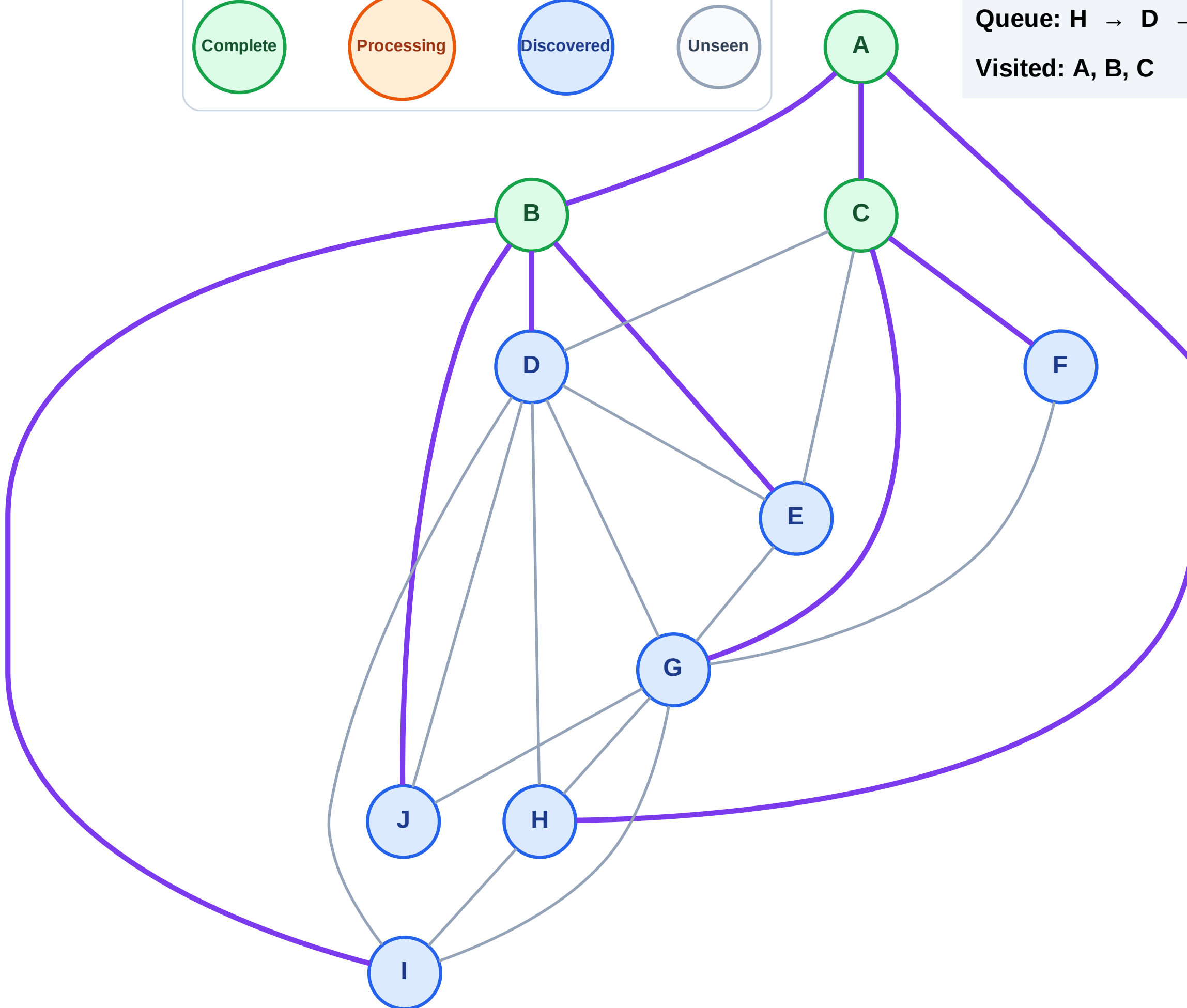
Step 7: Discover F, G from C

Legend



Queue: H → D → E → I → J → F → G

Visited: A, B, C



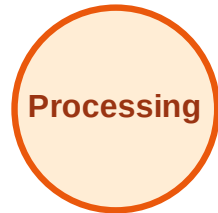
BFS Traversal

Step 8: Process node H

Legend



Complete



Processing



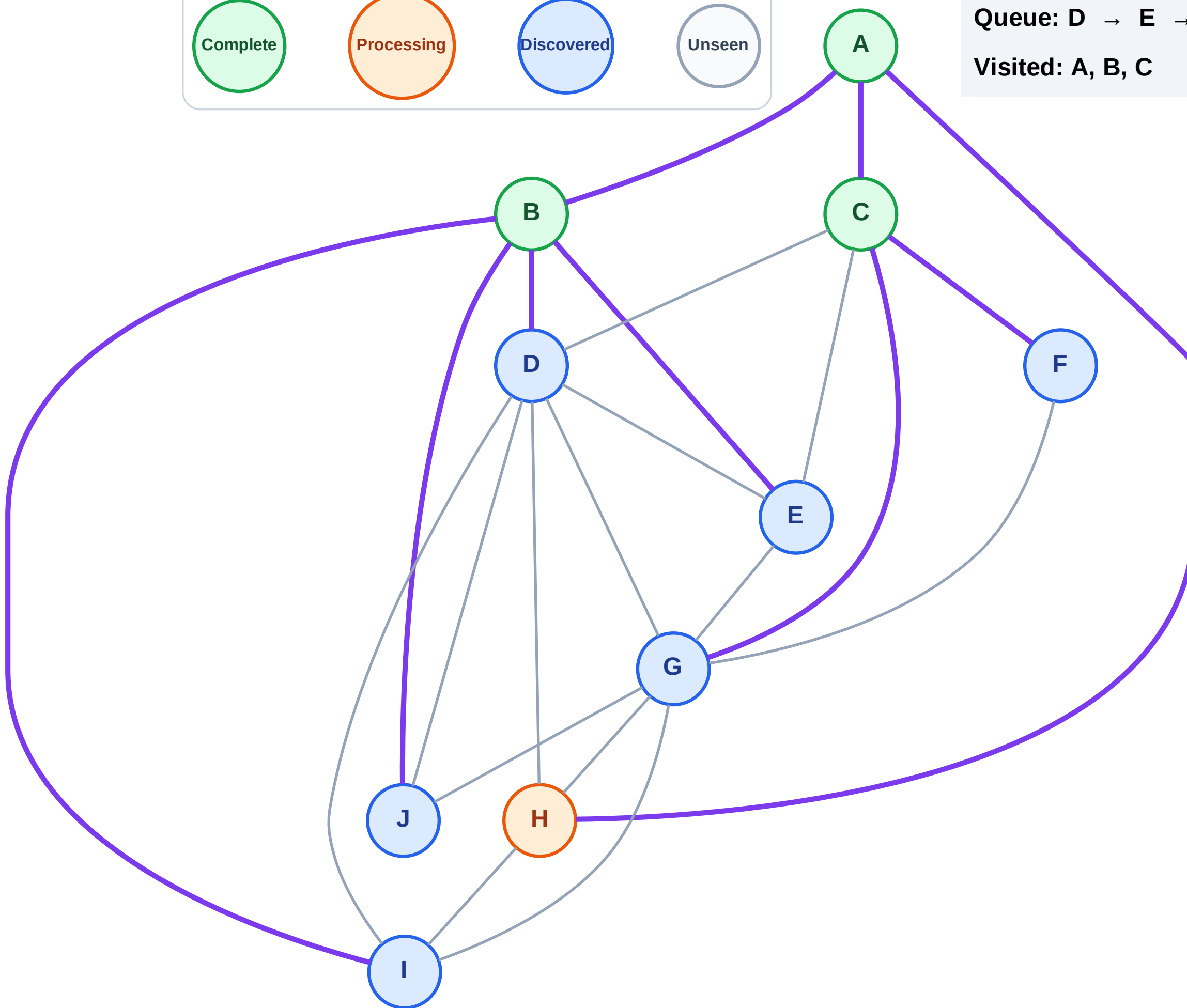
Discovered



Unseen

Queue: D → E → I → J → F → G

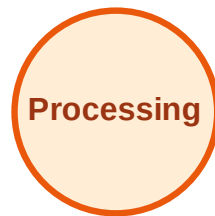
Visited: A, B, C



BFS Traversal

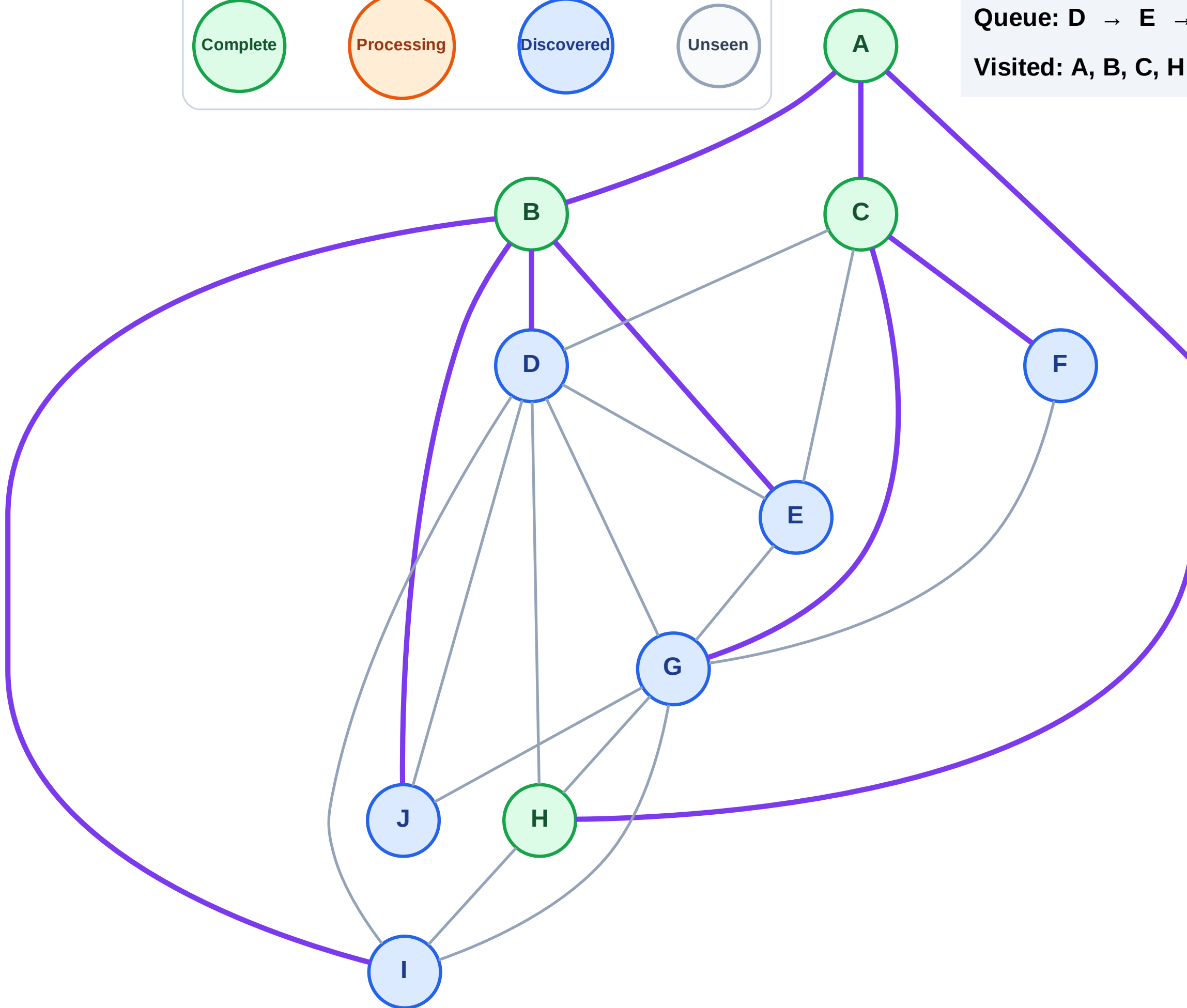
Step 9: No new neighbors from H

Legend



Queue: D → E → I → J → F → G

Visited: A, B, C, H

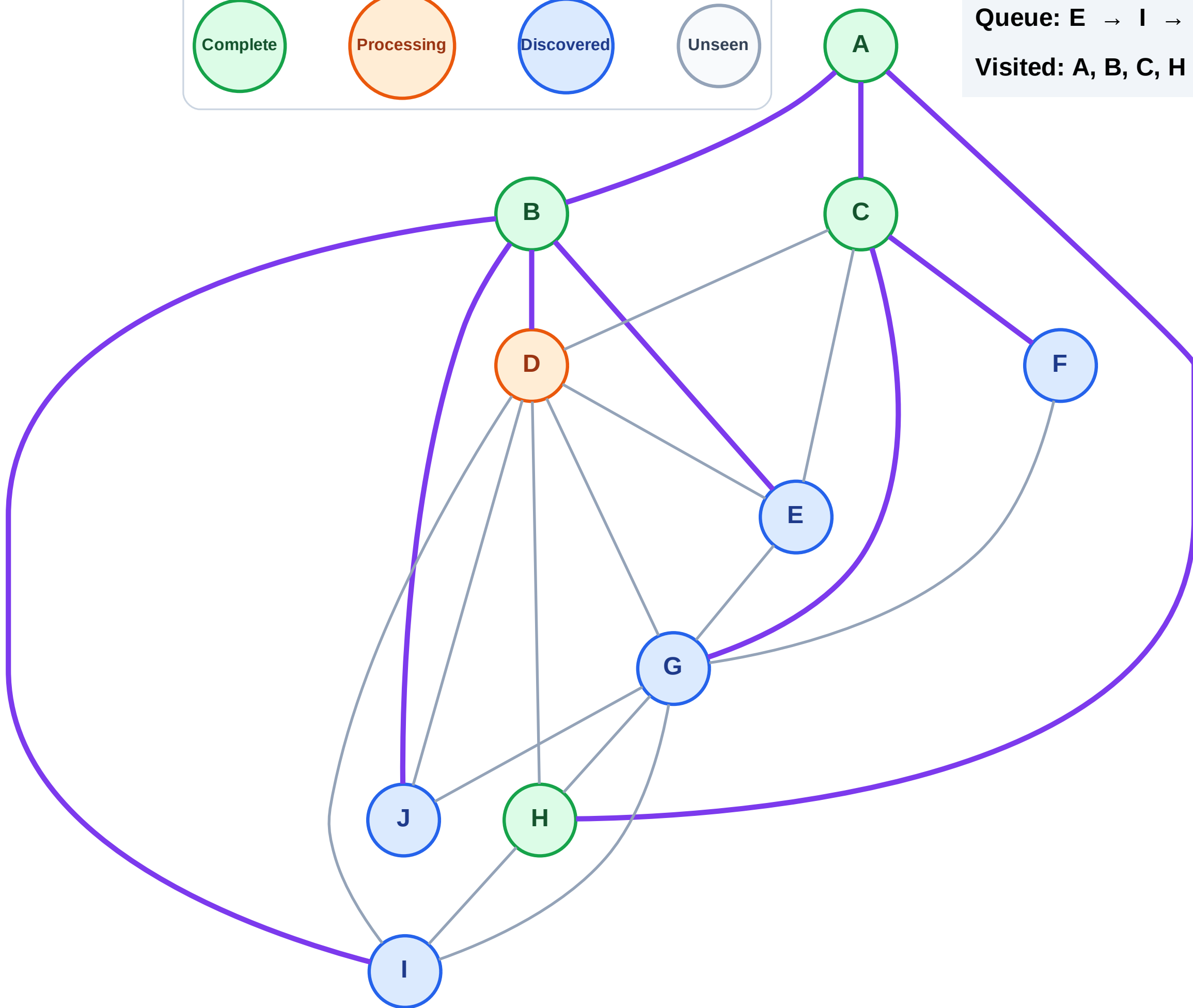


Step 10: Process node D

Legend

Unseen

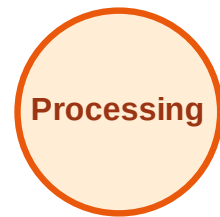
Visited: A, B, C, H



BFS Traversal

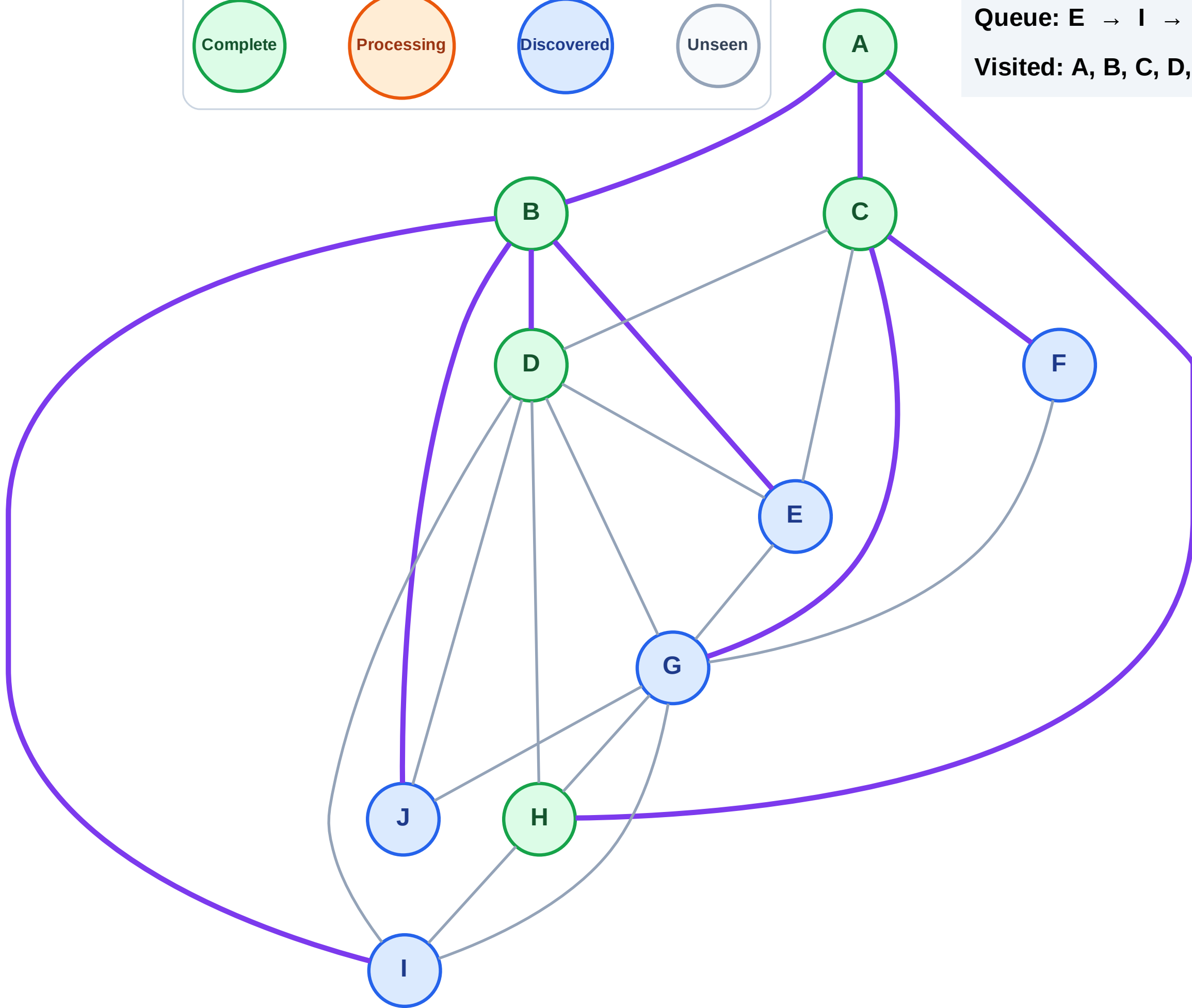
Step 11: No new neighbors from D

Legend



Queue: E → I → J → F → G

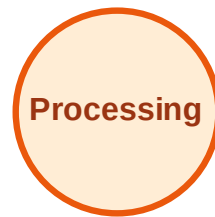
Visited: A, B, C, D, H



BFS Traversal

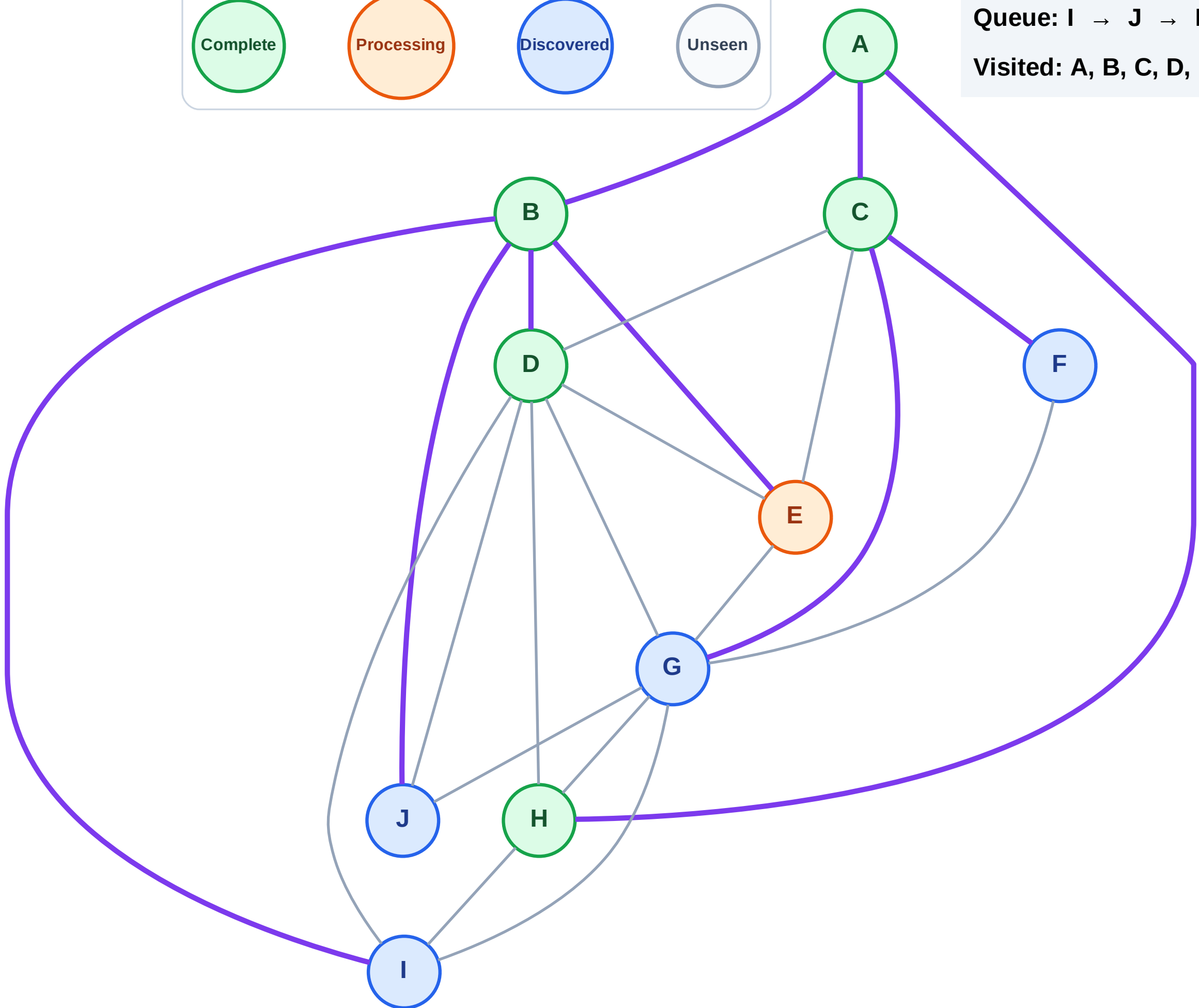
Step 12: Process node E

Legend



Queue: I → J → F → G

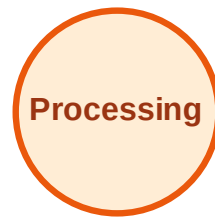
Visited: A, B, C, D, H



BFS Traversal

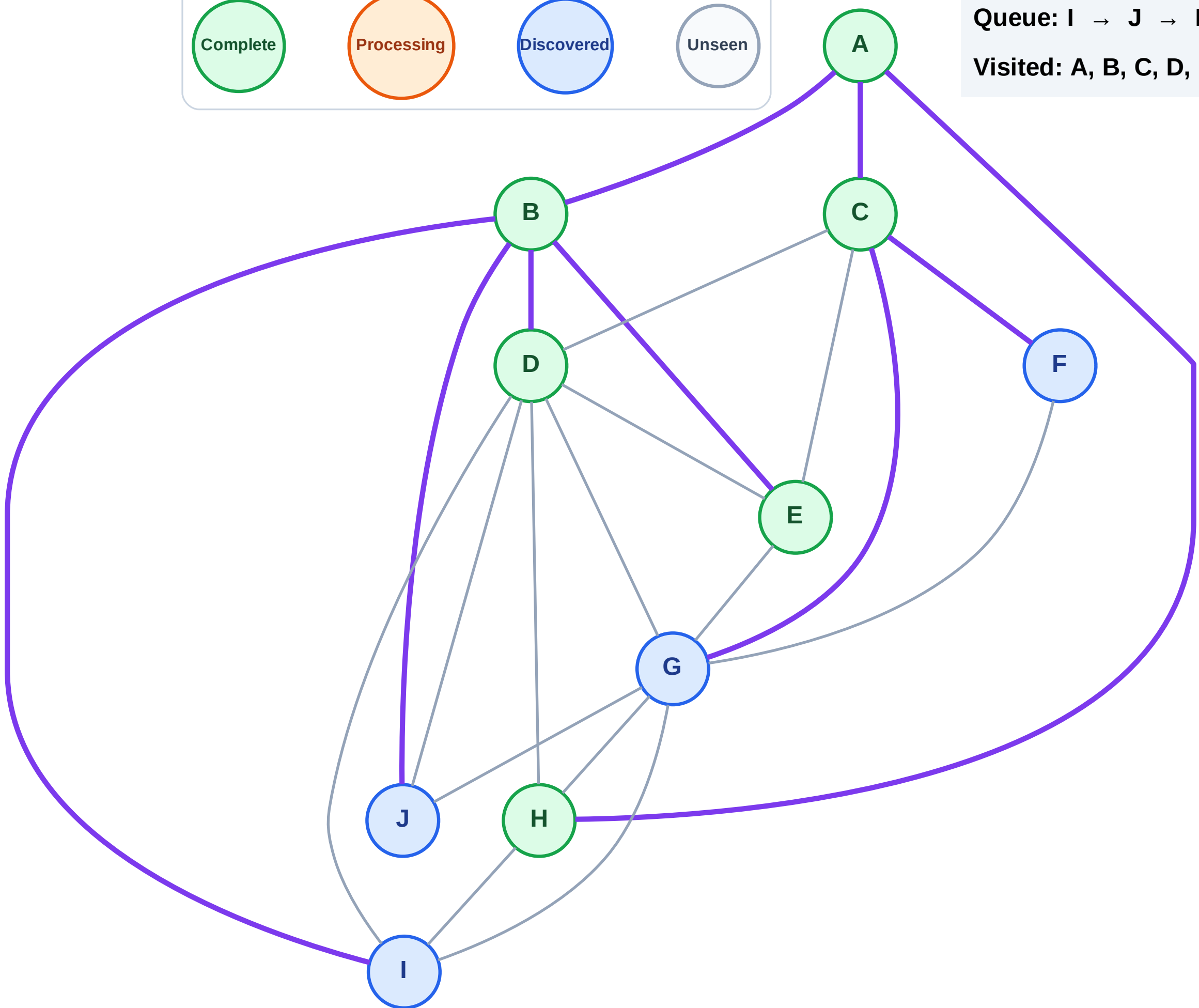
Step 13: No new neighbors from E

Legend



Queue: I → J → F → G

Visited: A, B, C, D, E, H



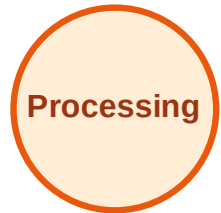
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



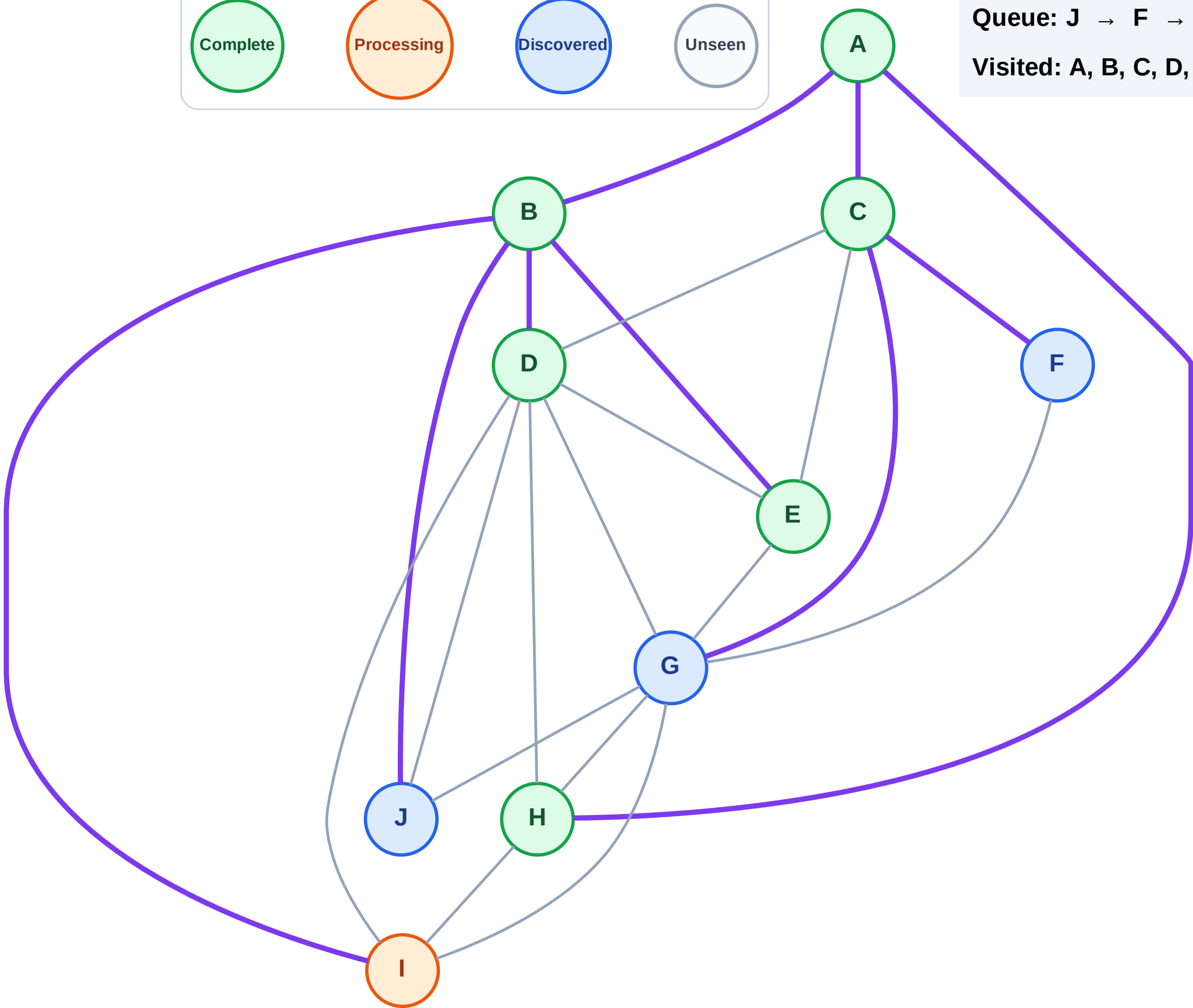
Discovered



Unseen

Queue: J → F → G

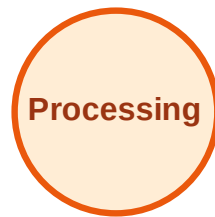
Visited: A, B, C, D, E, H



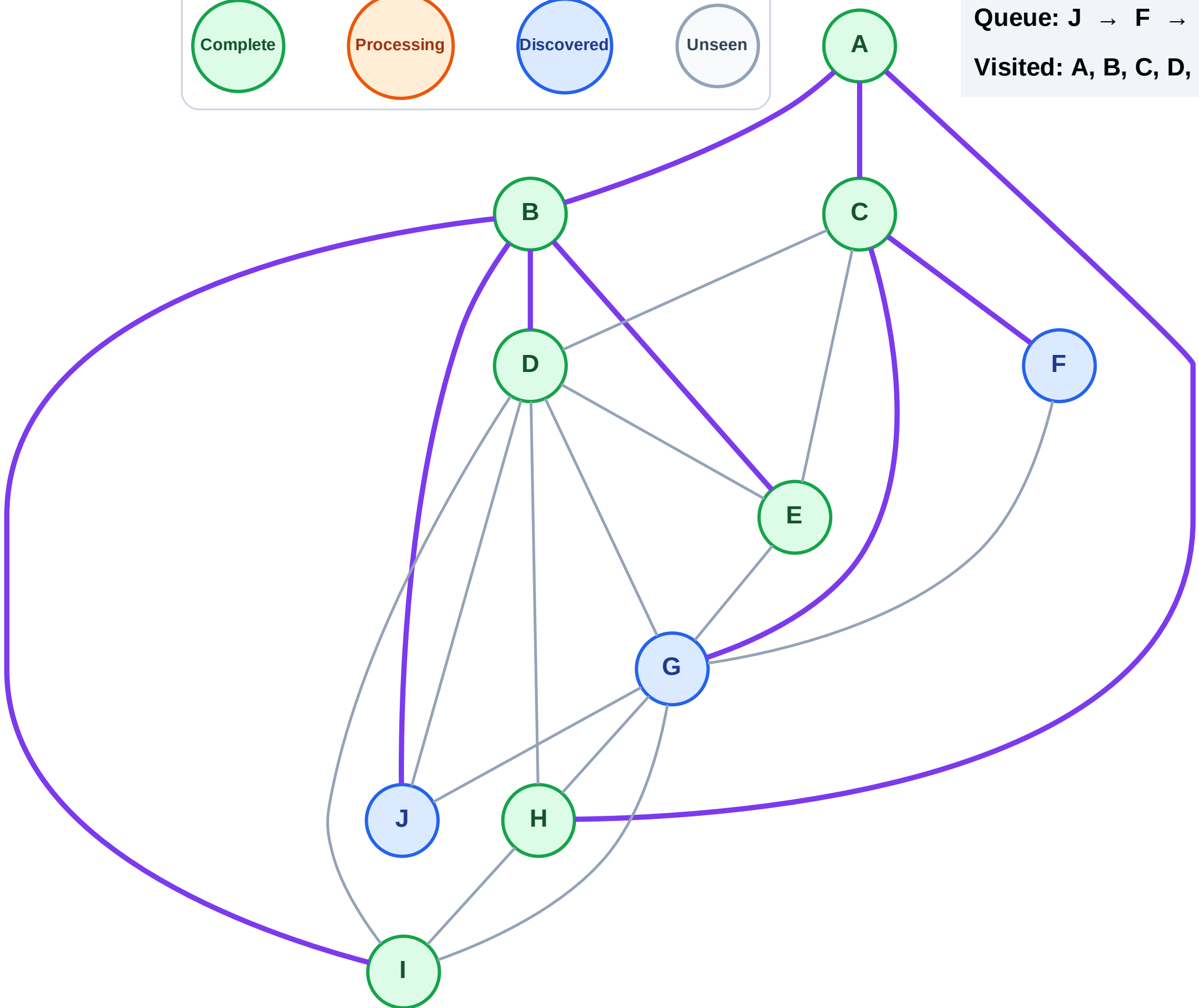
BFS Traversal

Step 15: No new neighbors from I

Legend



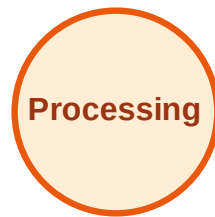
Queue: J → F → G
Visited: A, B, C, D, E, H, I



BFS Traversal

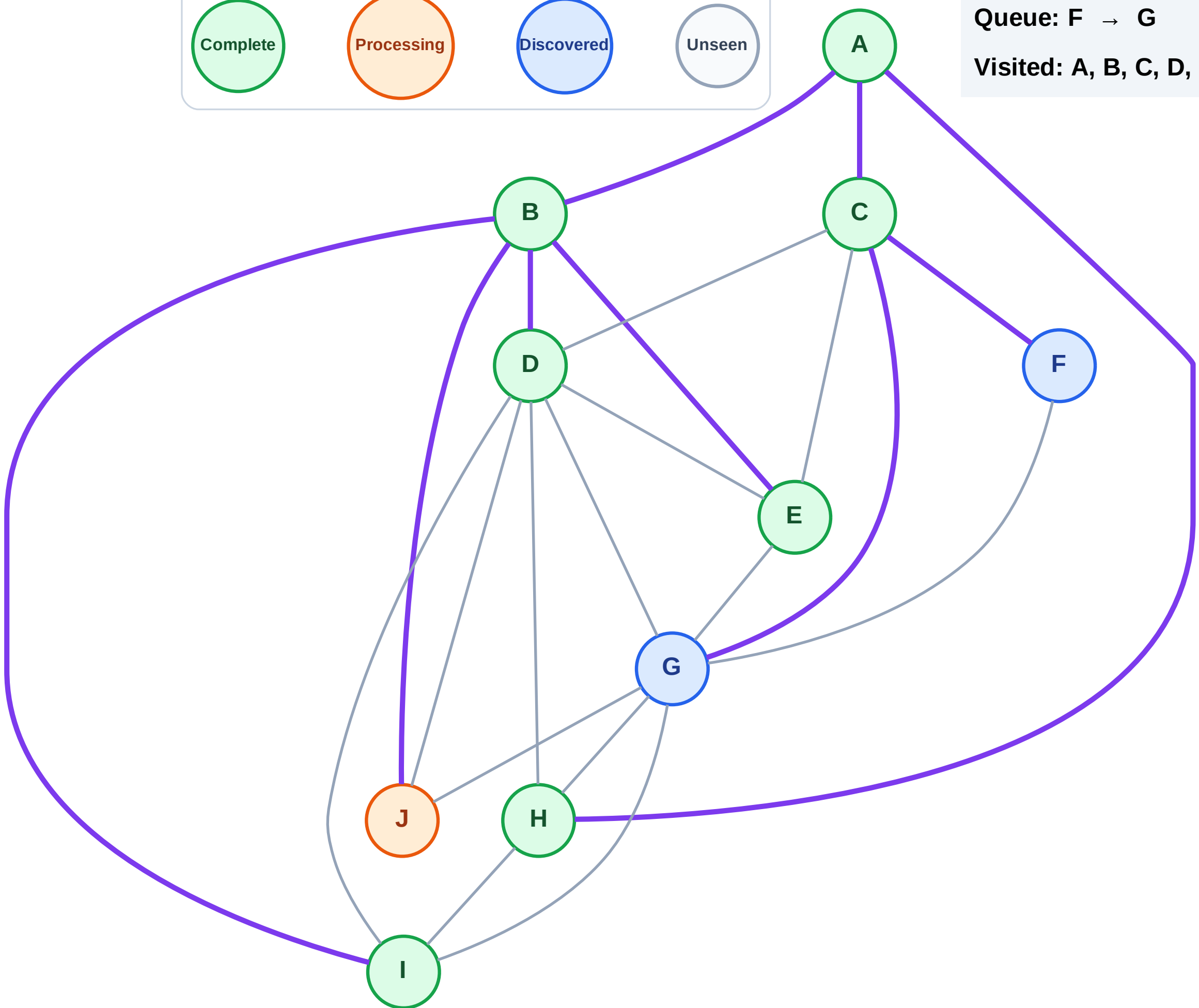
Step 16: Process node J

Legend



Queue: F → G

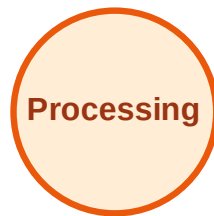
Visited: A, B, C, D, E, H, I



BFS Traversal

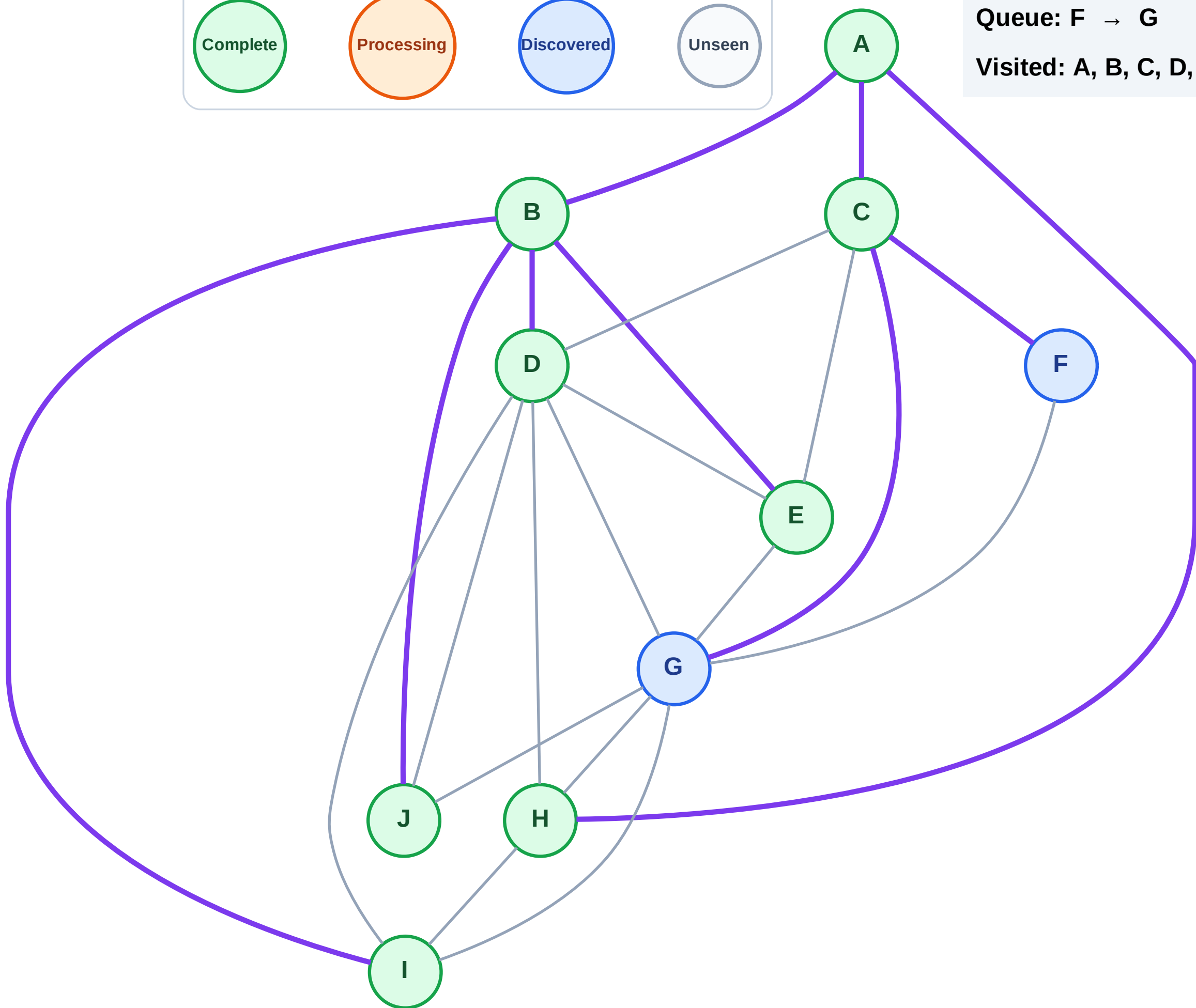
Step 17: No new neighbors from J

Legend



Queue: F → G

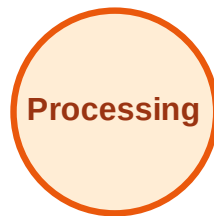
Visited: A, B, C, D, E, H, I, J



BFS Traversal

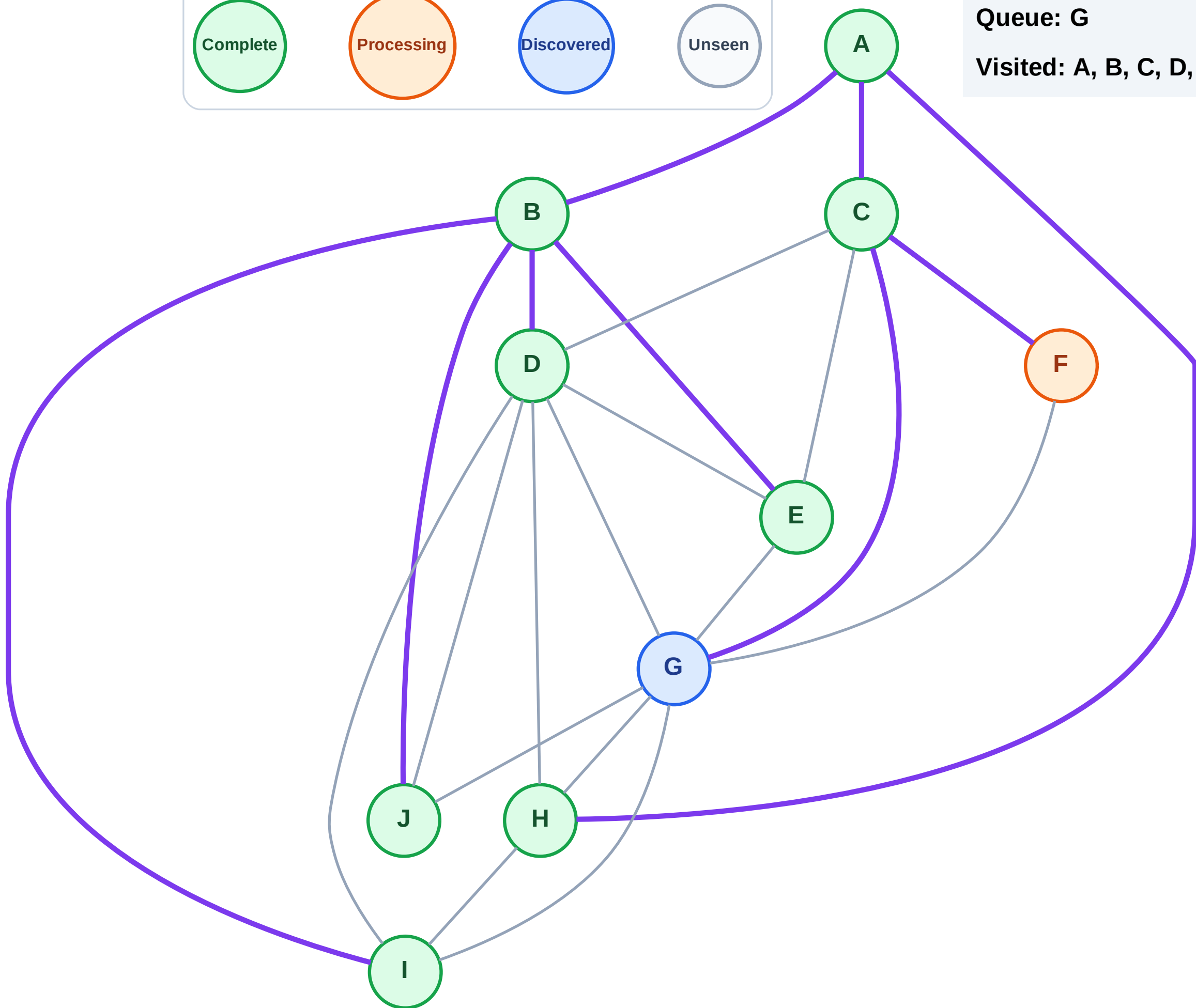
Step 18: Process node F

Legend



Queue: G

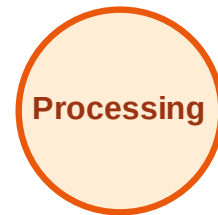
Visited: A, B, C, D, E, H, I, J



BFS Traversal

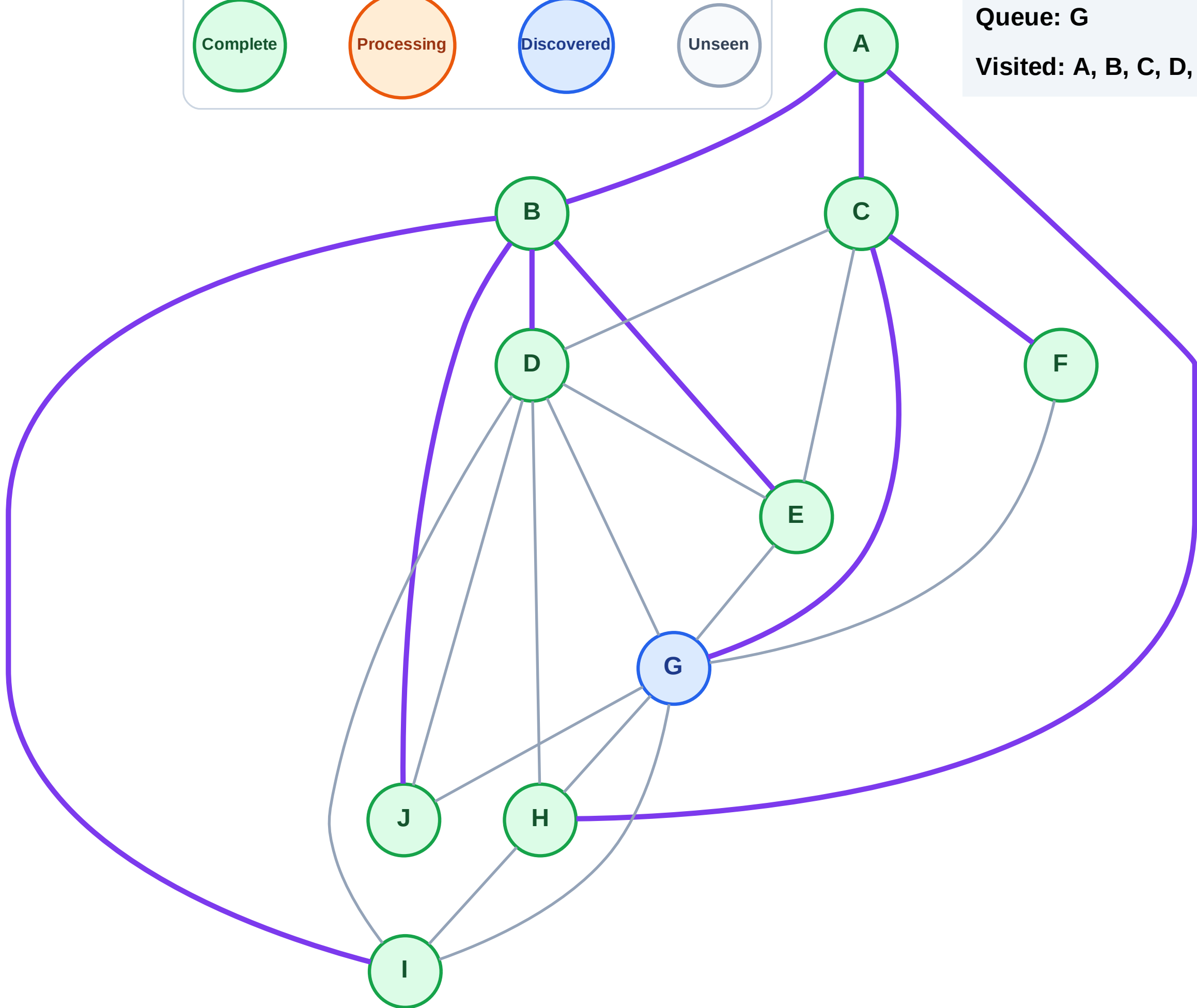
Step 19: No new neighbors from F

Legend



Queue: G

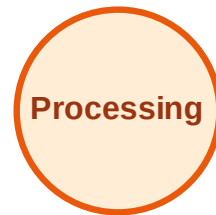
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

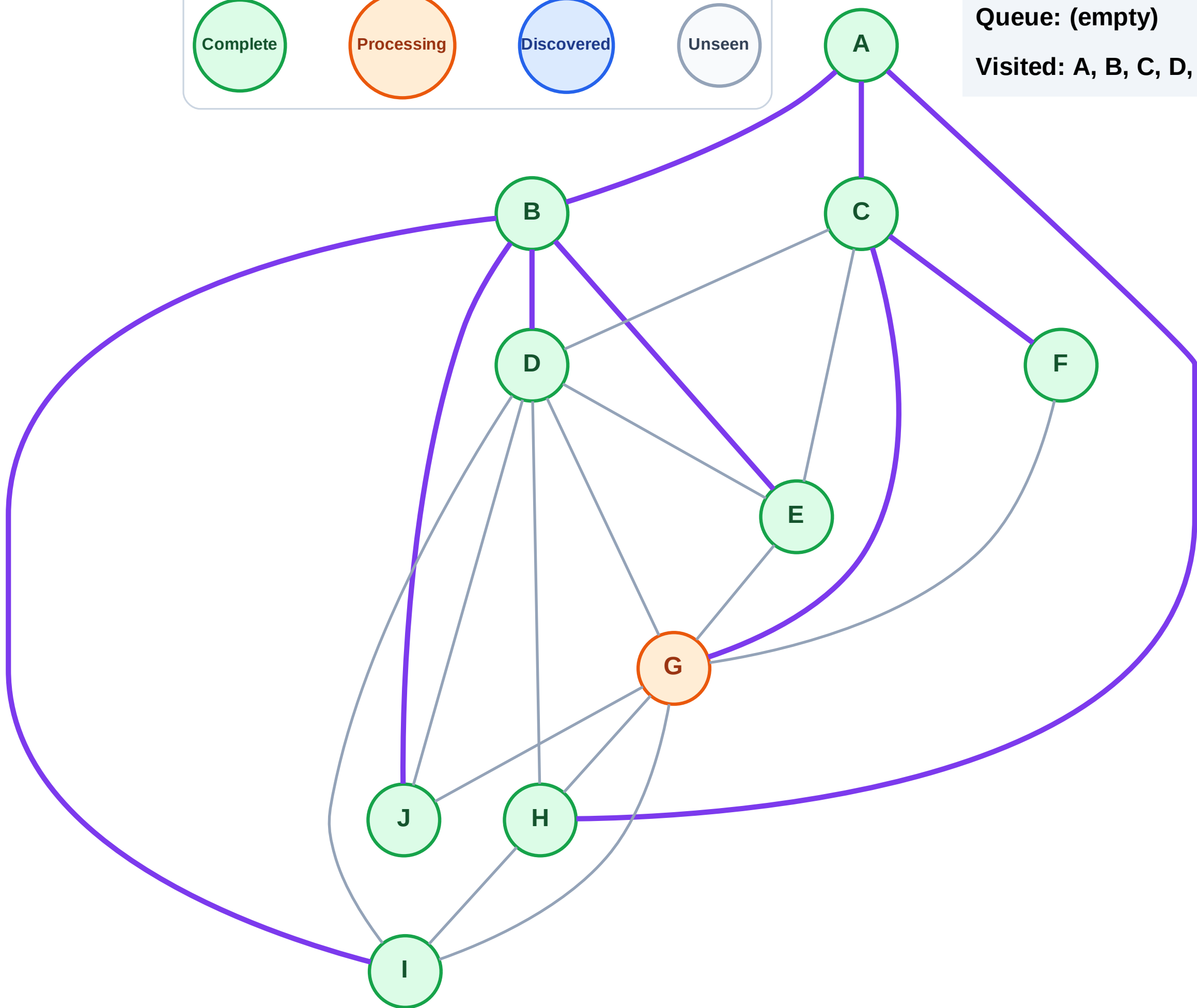
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J

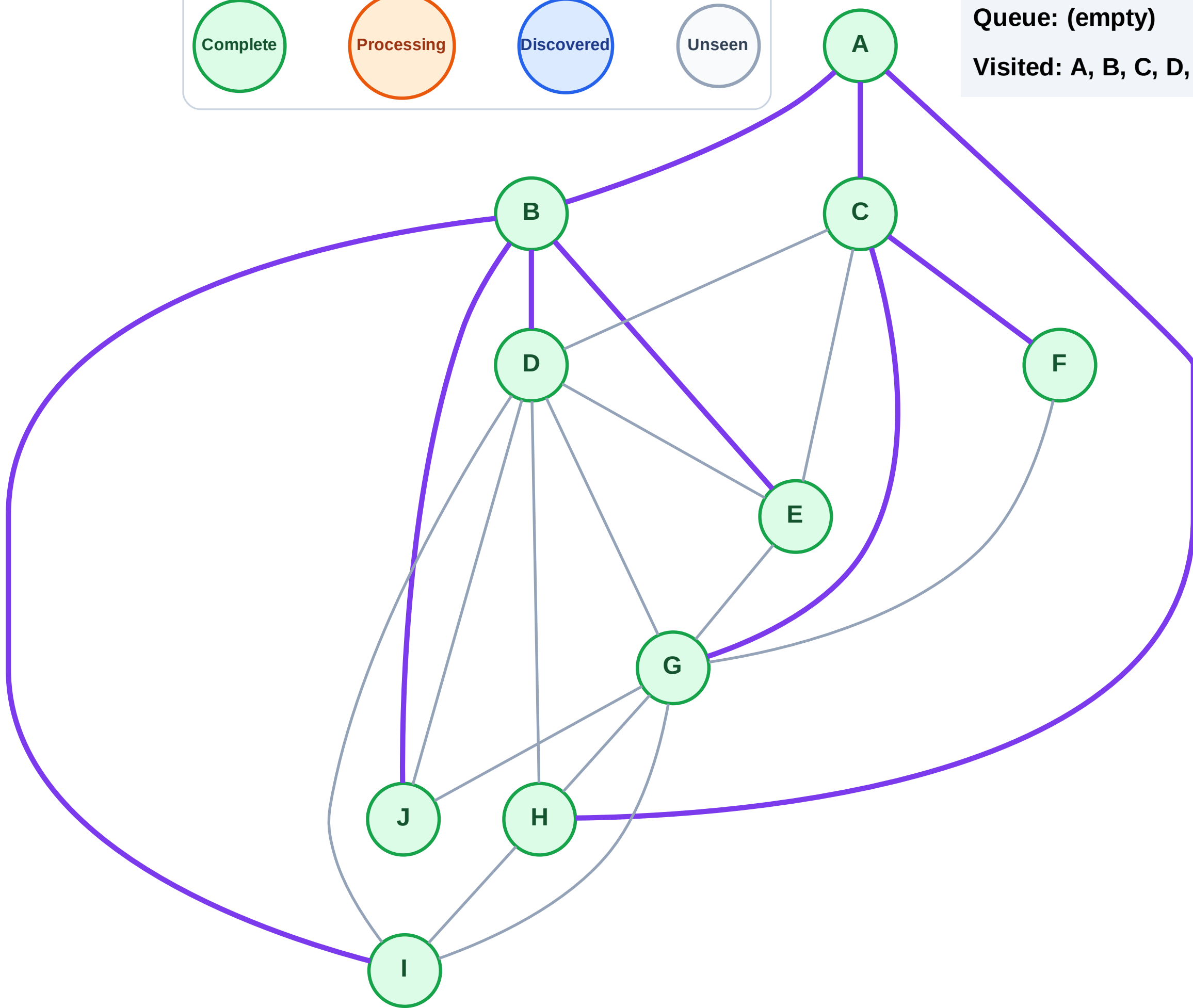


Step 21: No new neighbors from G

Legend



Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

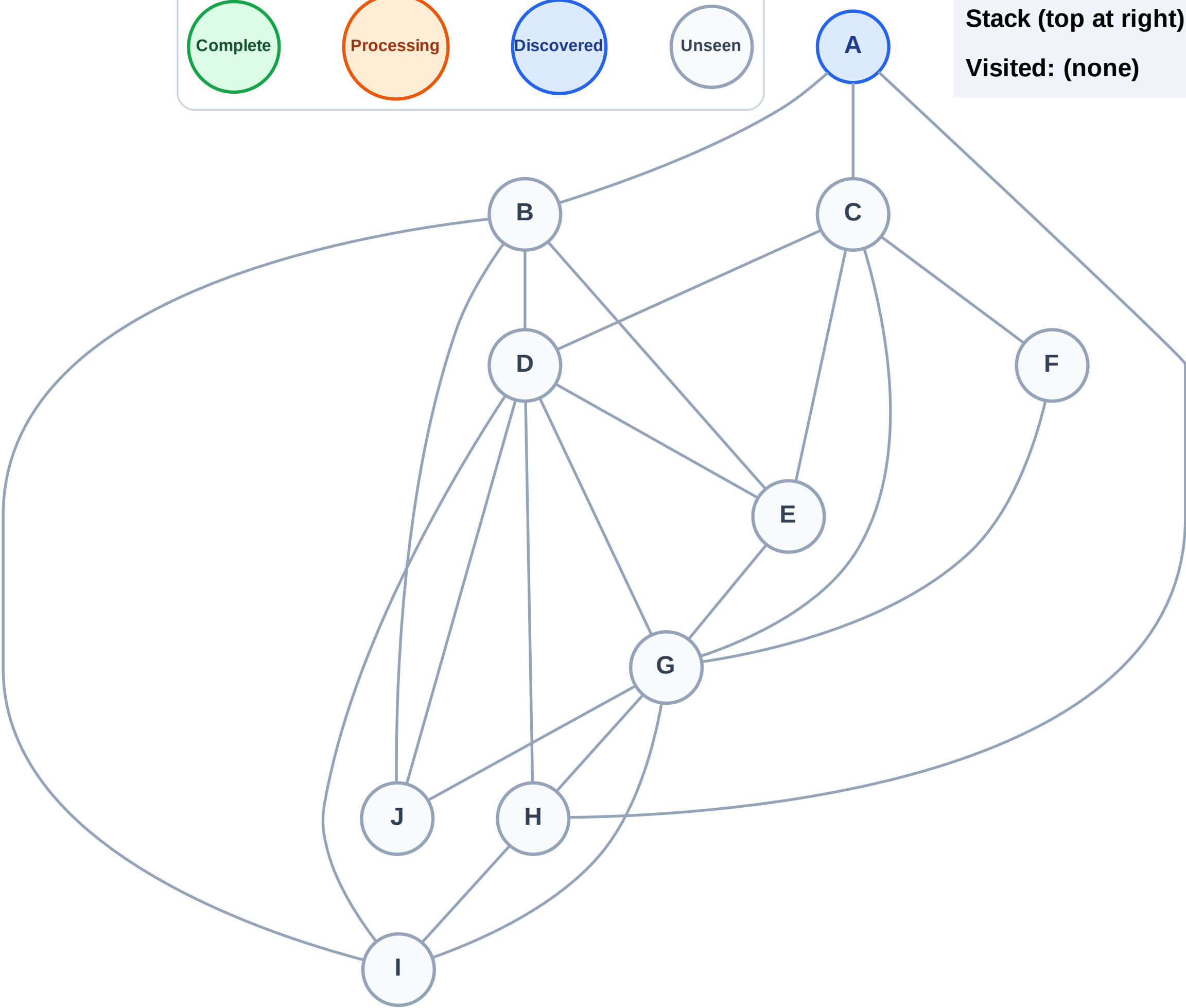
Step 1: Start at A

Legend



Stack (top at right): A

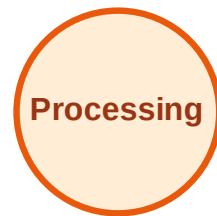
Visited: (none)



DFS Traversal

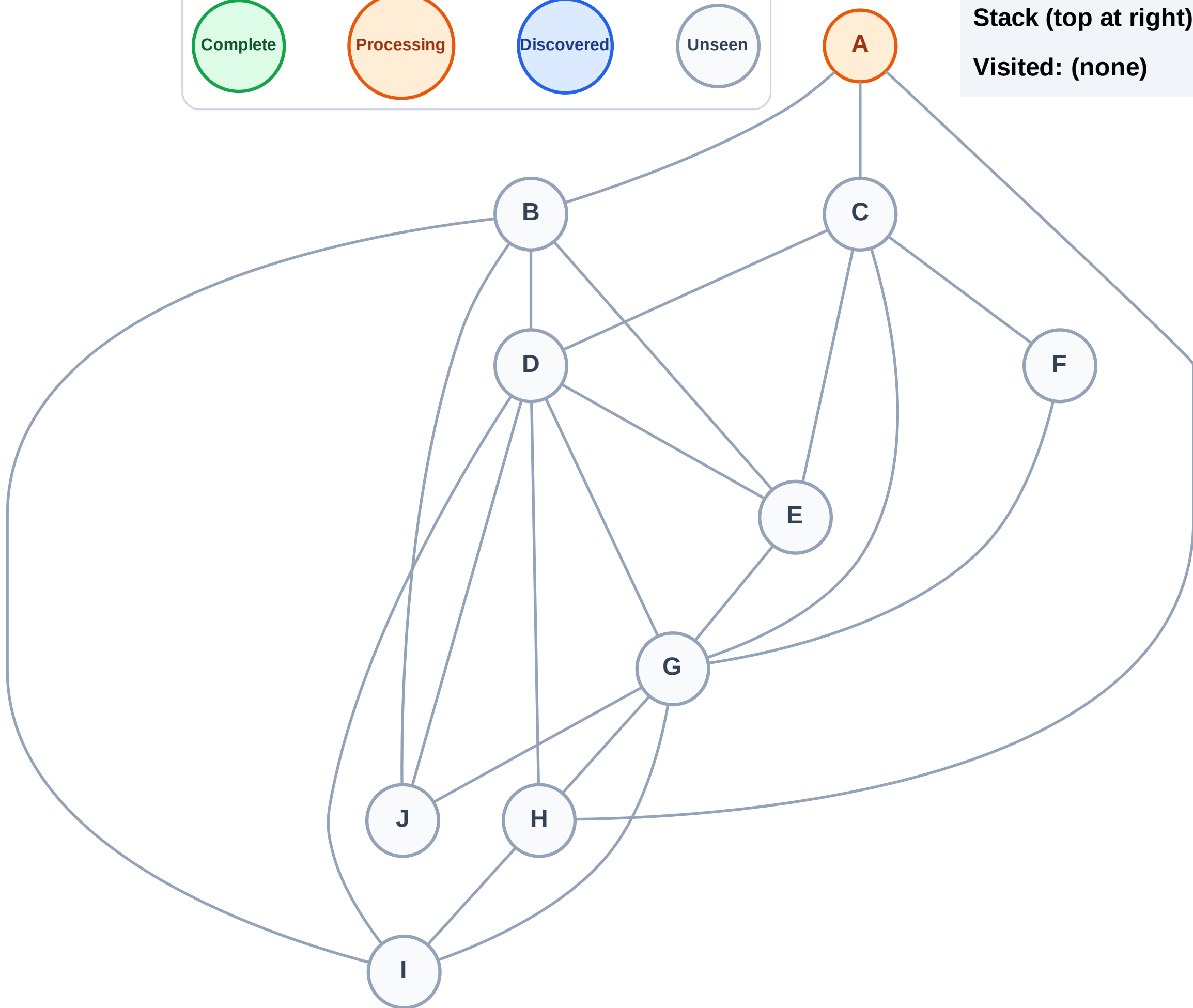
Step 2: Process node A

Legend



Stack (top at right): (empty)

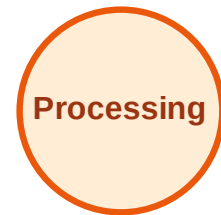
Visited: (none)



DFS Traversal

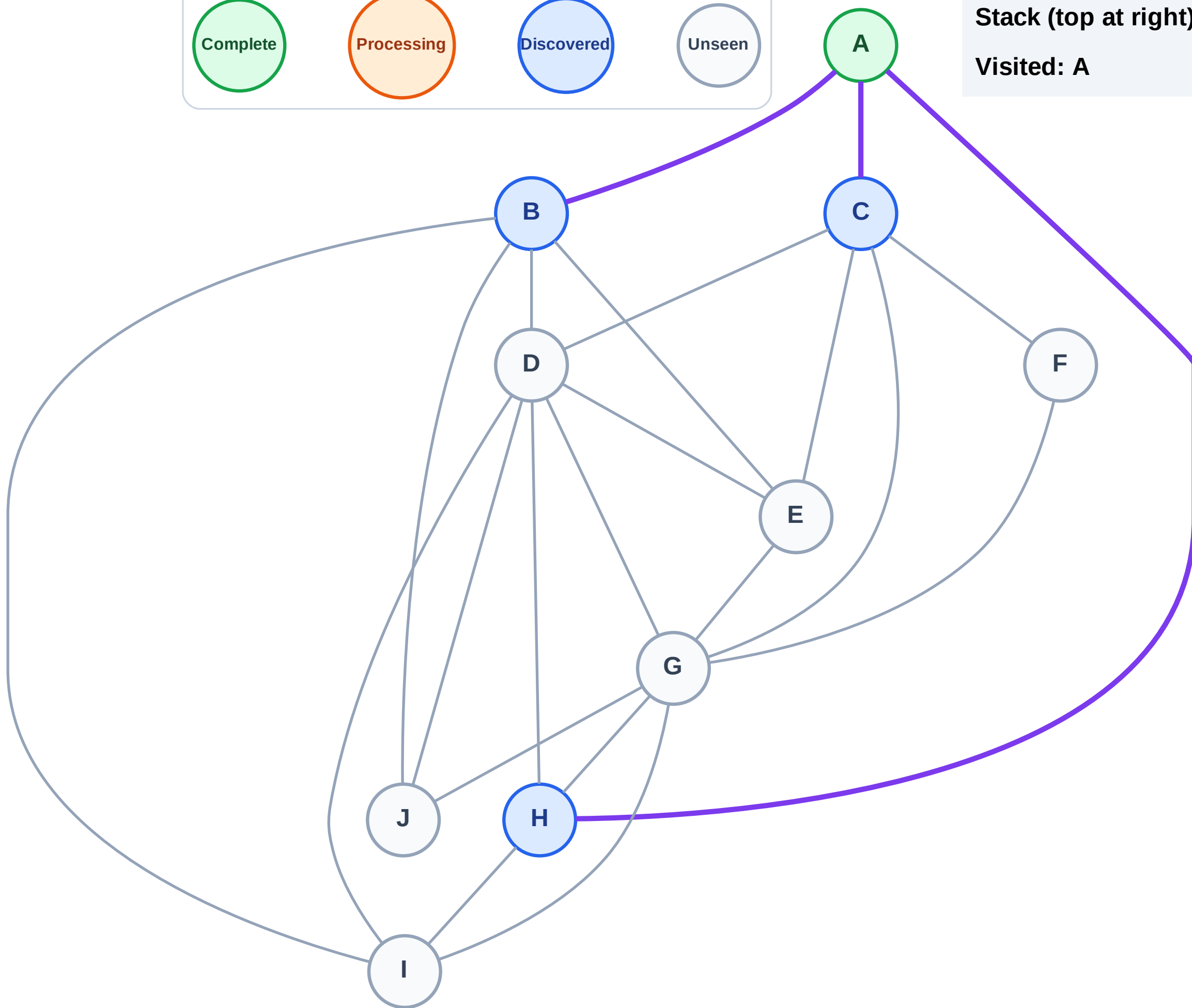
Step 3: Discover H, C, B from A

Legend



Stack (top at right): H | C | B

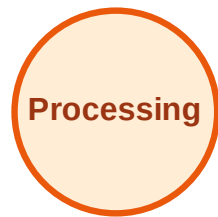
Visited: A



DFS Traversal

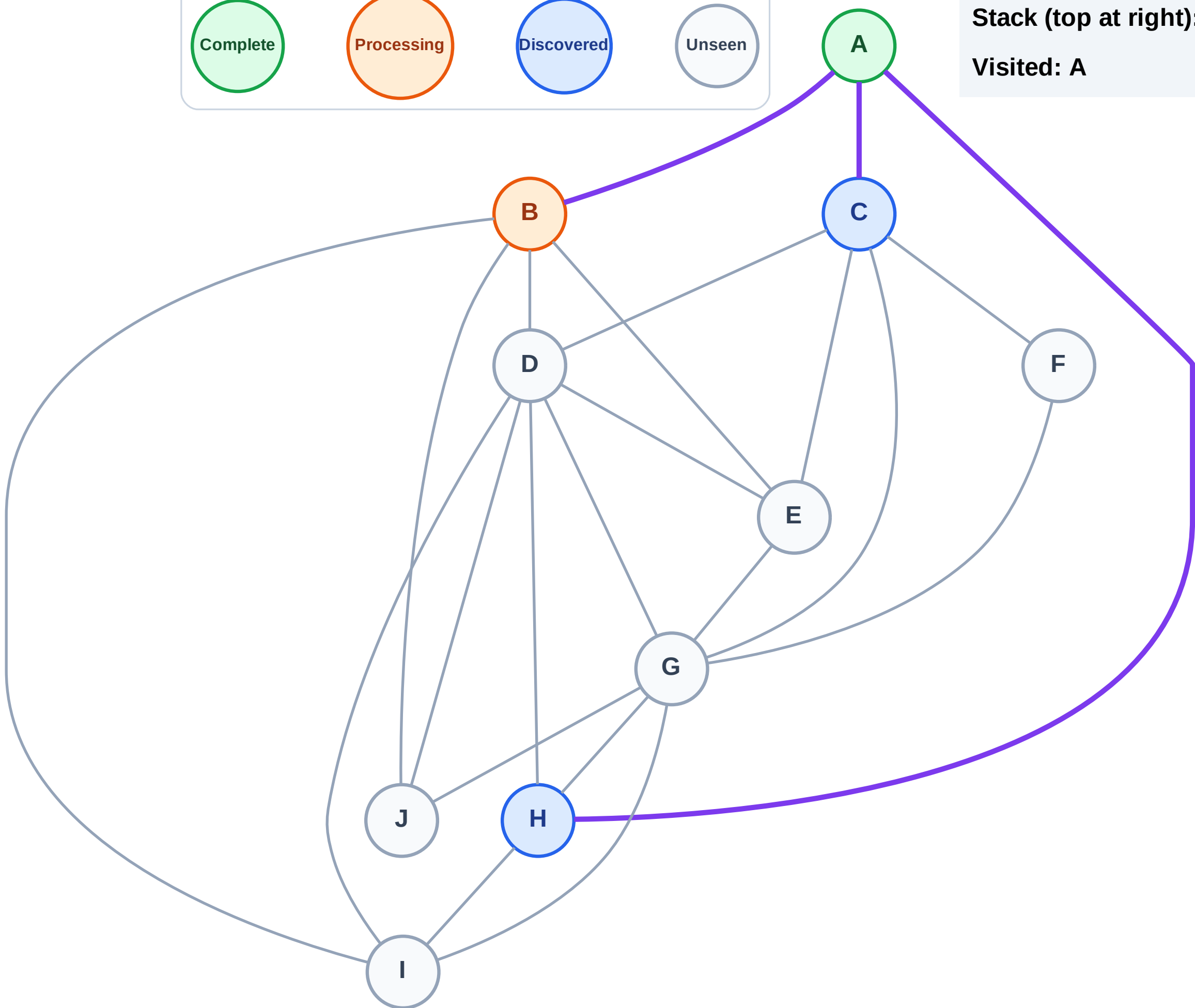
Step 4: Process node B

Legend



Stack (top at right): H | C

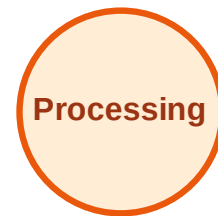
Visited: A



DFS Traversal

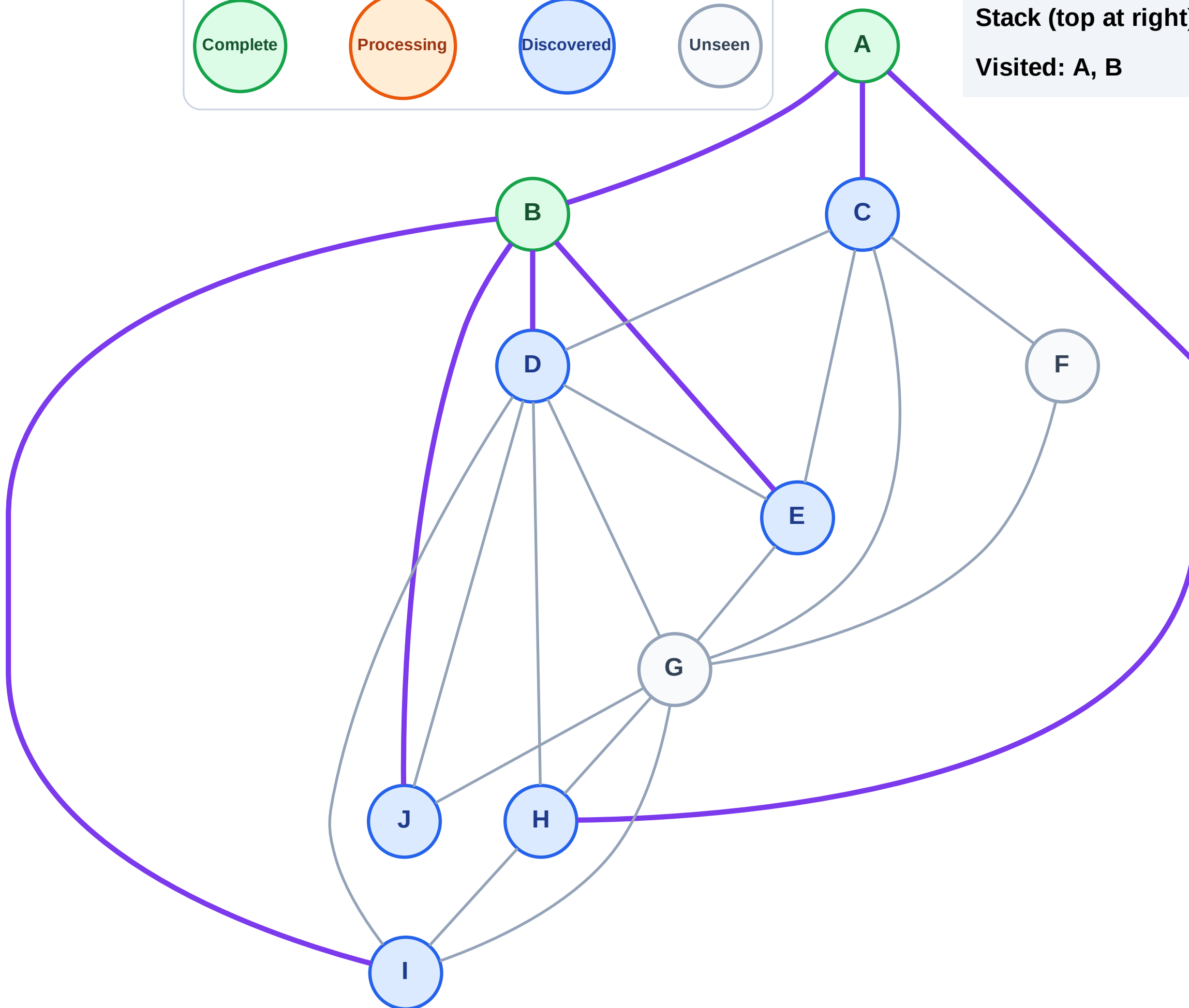
Step 5: Discover J, I, E, D from B

Legend



Stack (top at right): H | C | J | I | E | D

Visited: A, B



Step 6: Process node D

Page 10 of 10

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

- Complete
- Processing
- Discovered
- Unseen

Complete

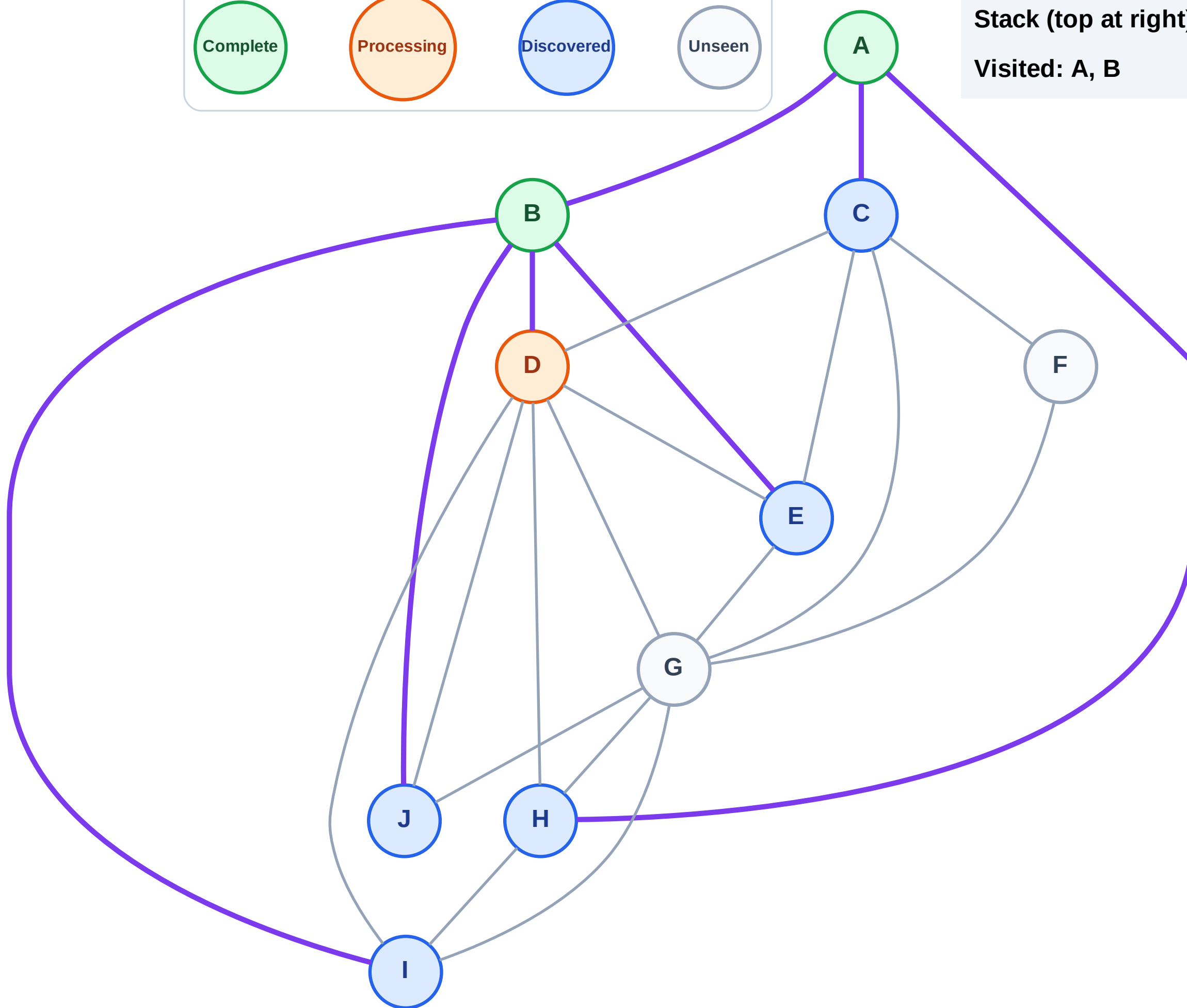
Processing

Discovered

Unseen

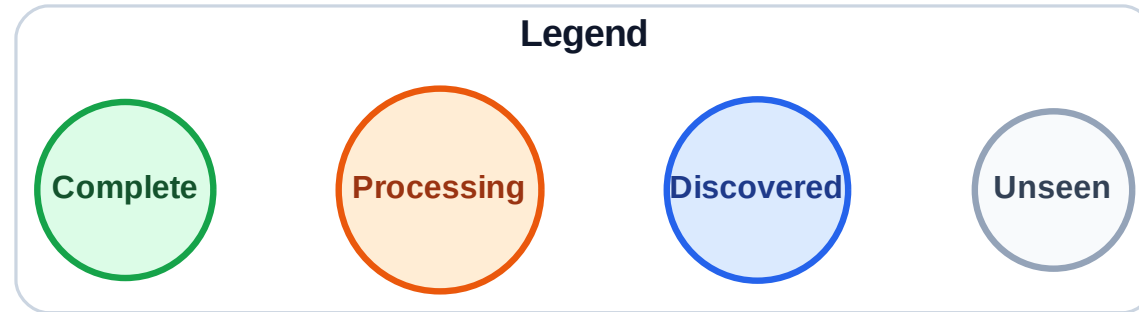
Stack (top at right): H | C | J | I | E

Visited: A, B



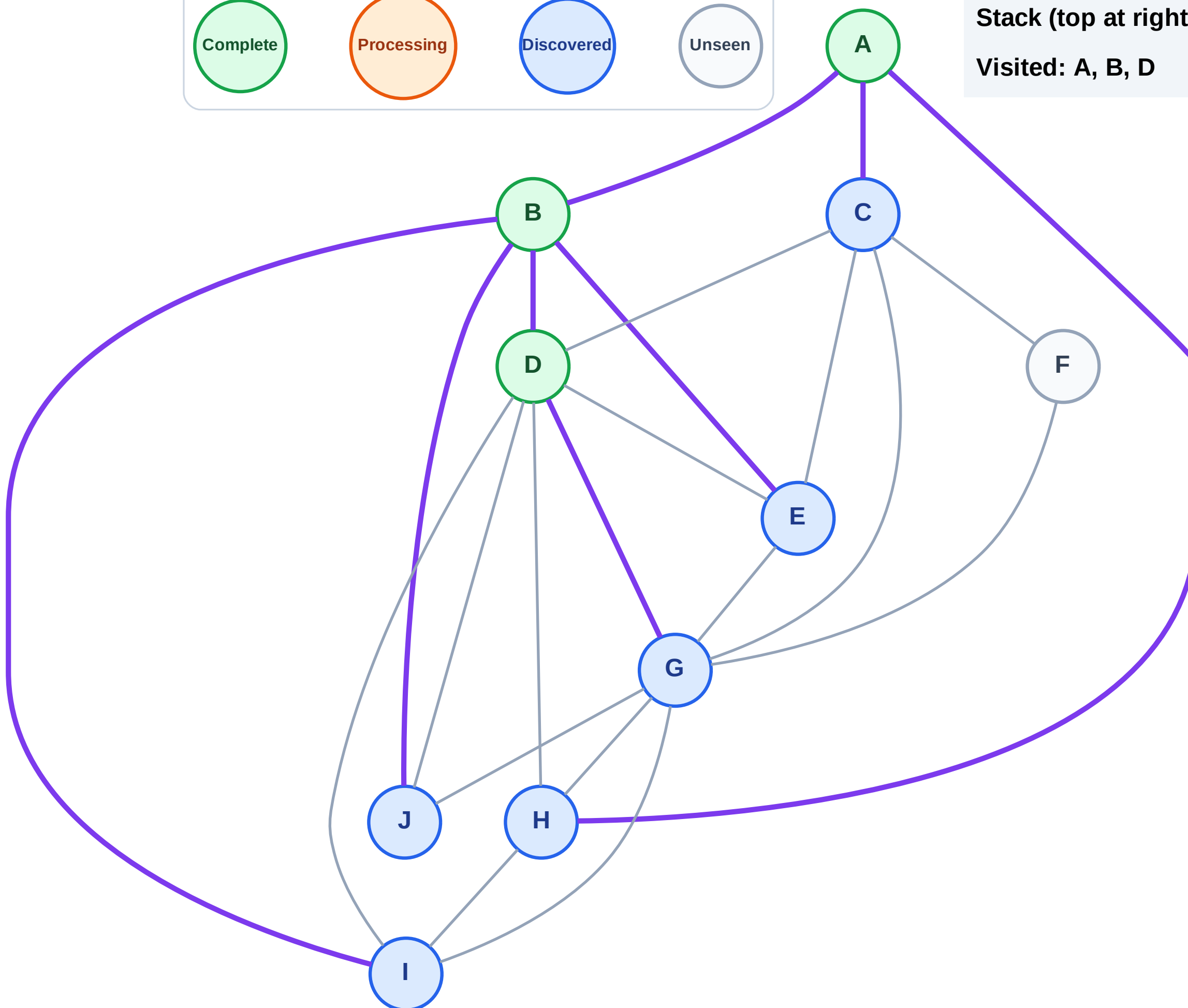
Step 7: Discover G from D

Step 7: Discover G from D



Stack (top at right): H | C | J | I | E | G

Visited: A, B, D



DFS Traversal

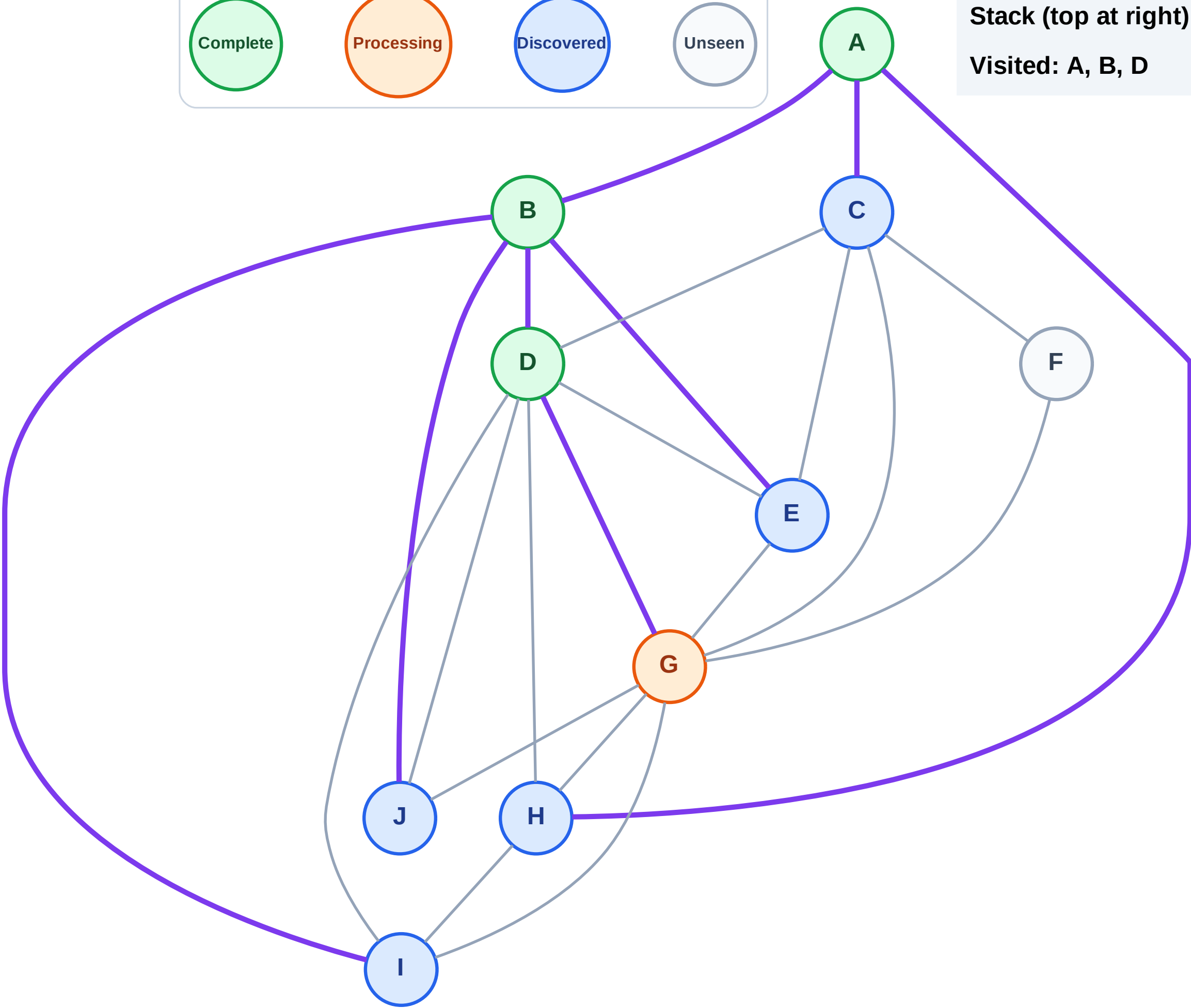
Step 8: Process node G

Legend



Stack (top at right): H | C | J | I | E

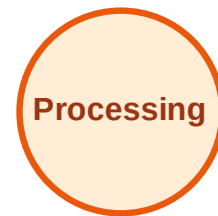
Visited: A, B, D



DFS Traversal

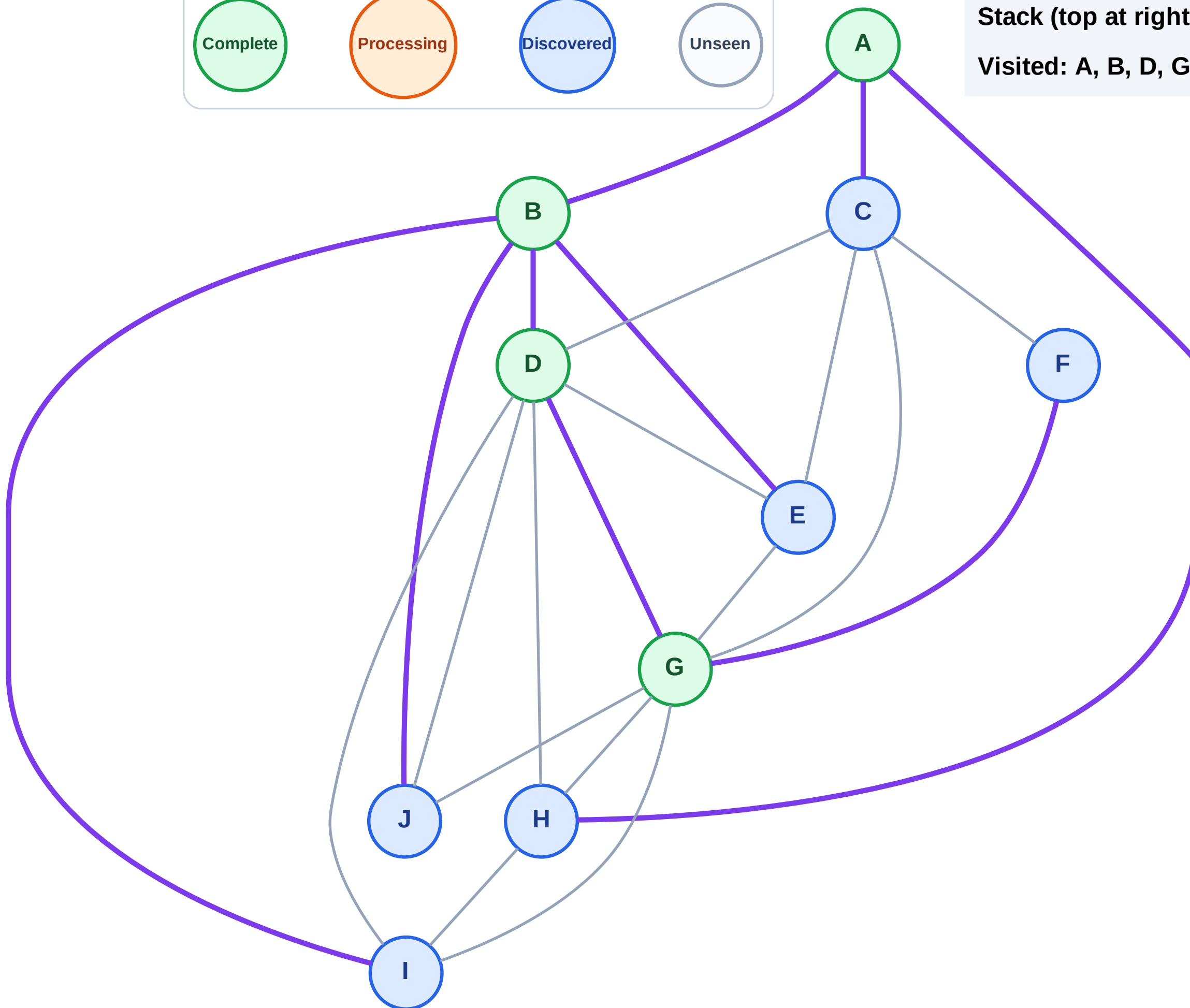
Step 9: Discover F from G

Legend



Stack (top at right): H | C | J | I | E | F

Visited: A, B, D, G



DFS Traversal

Step 10: Process node F

Legend

Complete

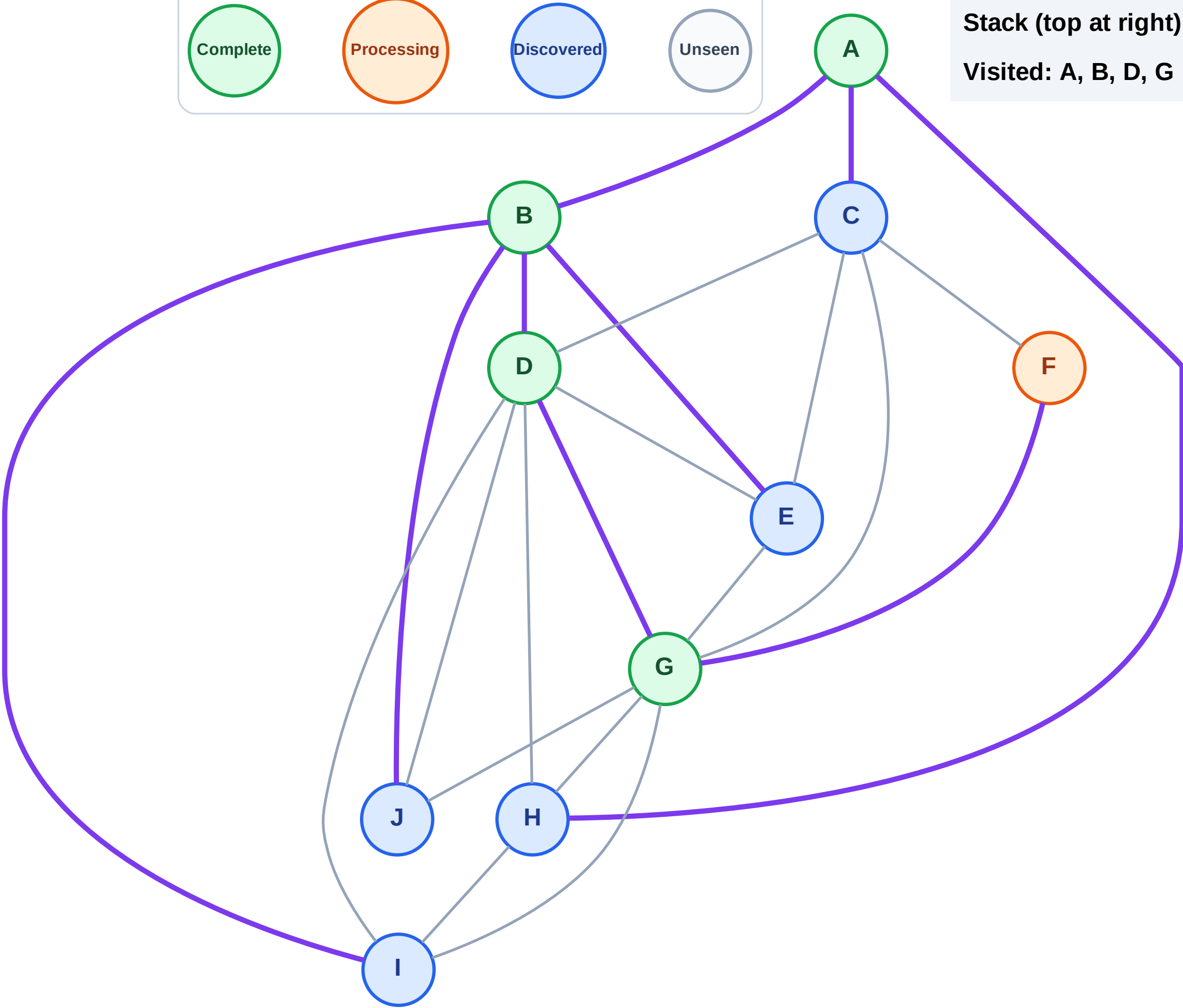
Processing

Discovered

Unseen

Stack (top at right): H | C | J | I | E

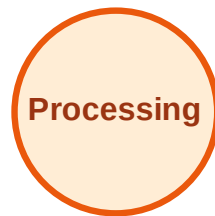
Visited: A, B, D, G



DFS Traversal

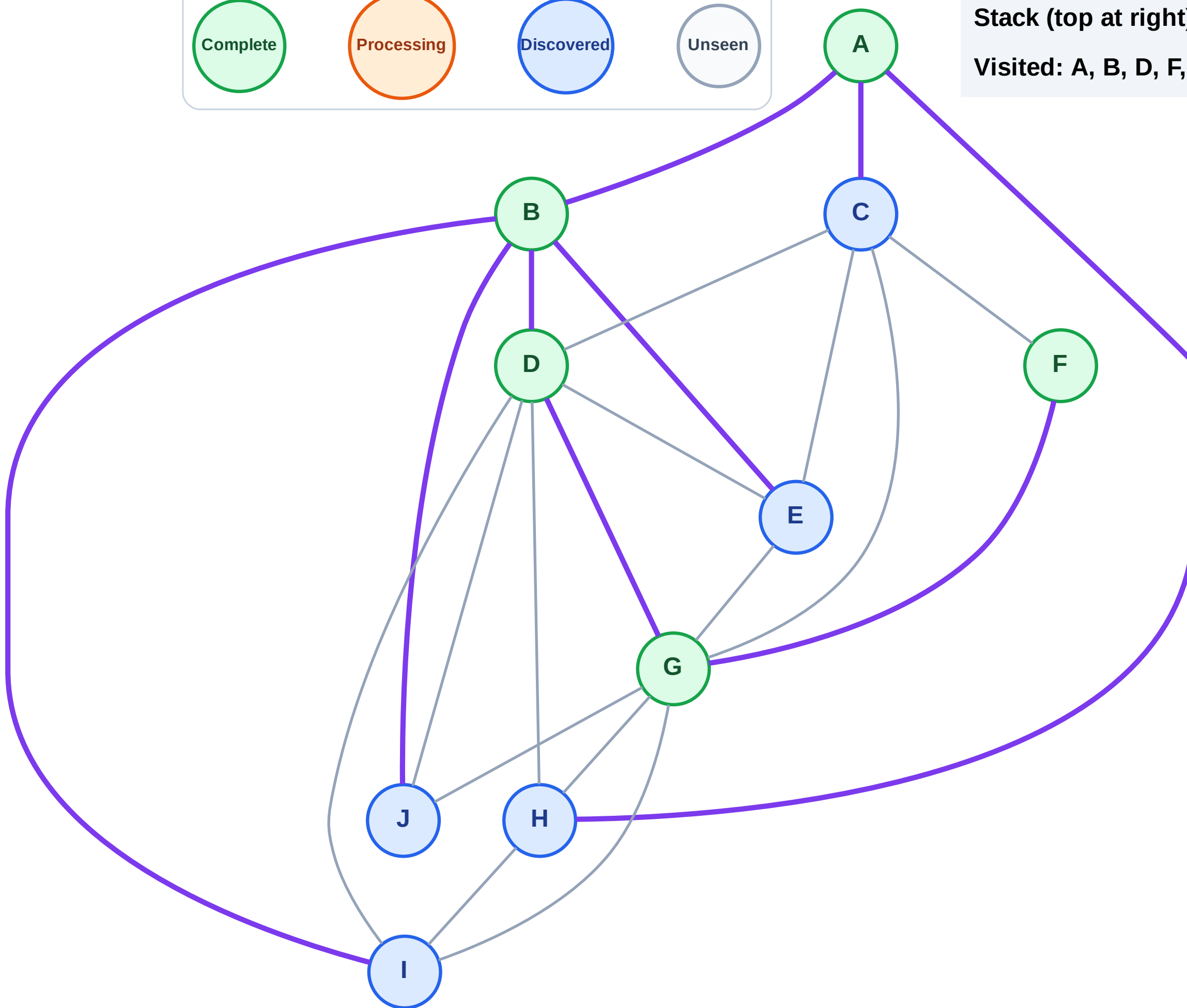
Step 11: No new neighbors from F

Legend



Stack (top at right): H | C | J | I | E

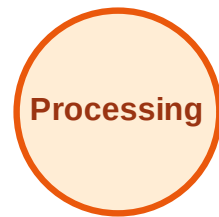
Visited: A, B, D, F, G



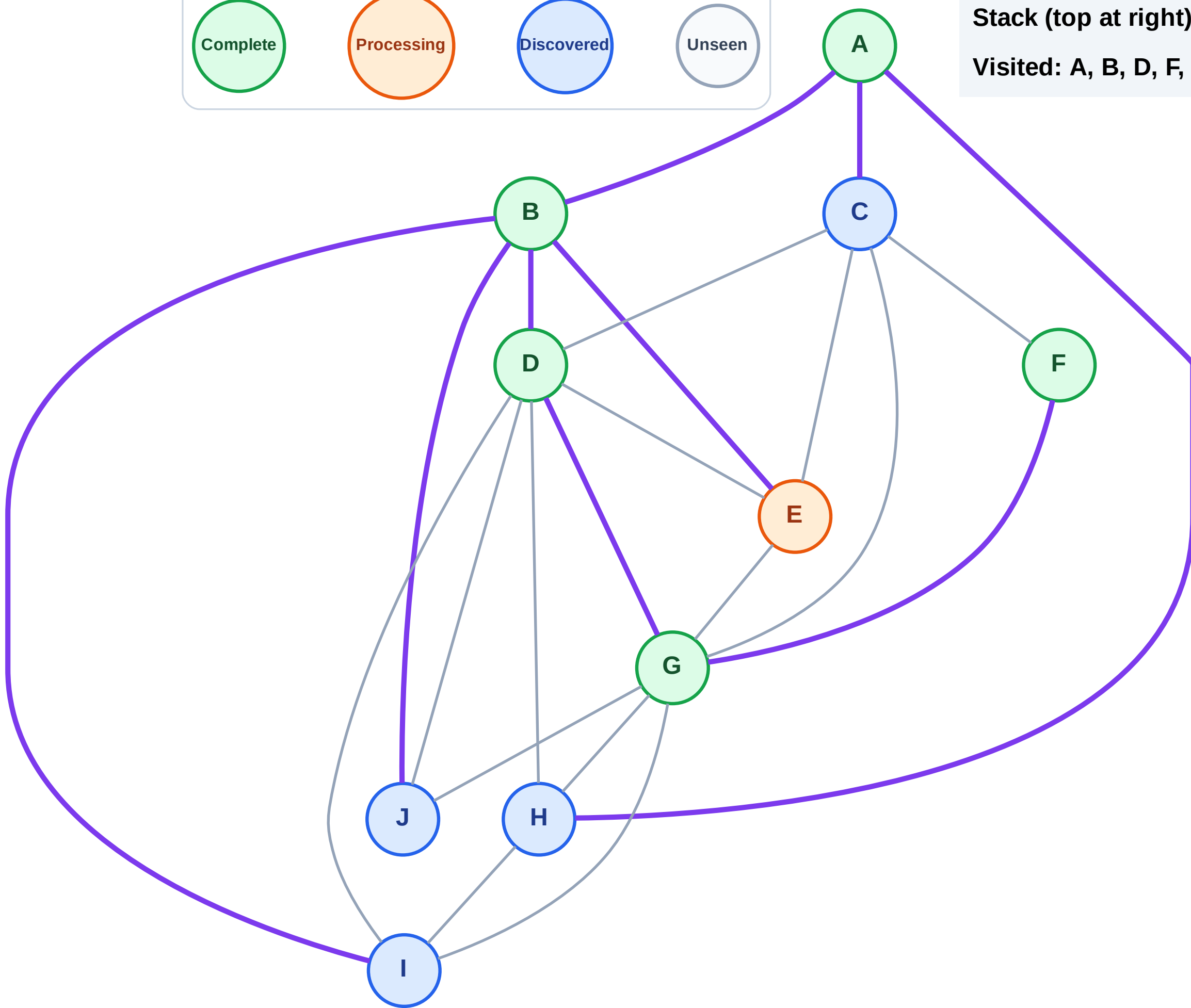
DFS Traversal

Step 12: Process node E

Legend



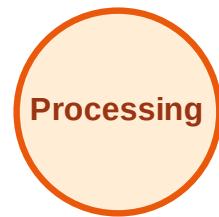
Stack (top at right): H | C | J | I
Visited: A, B, D, F, G



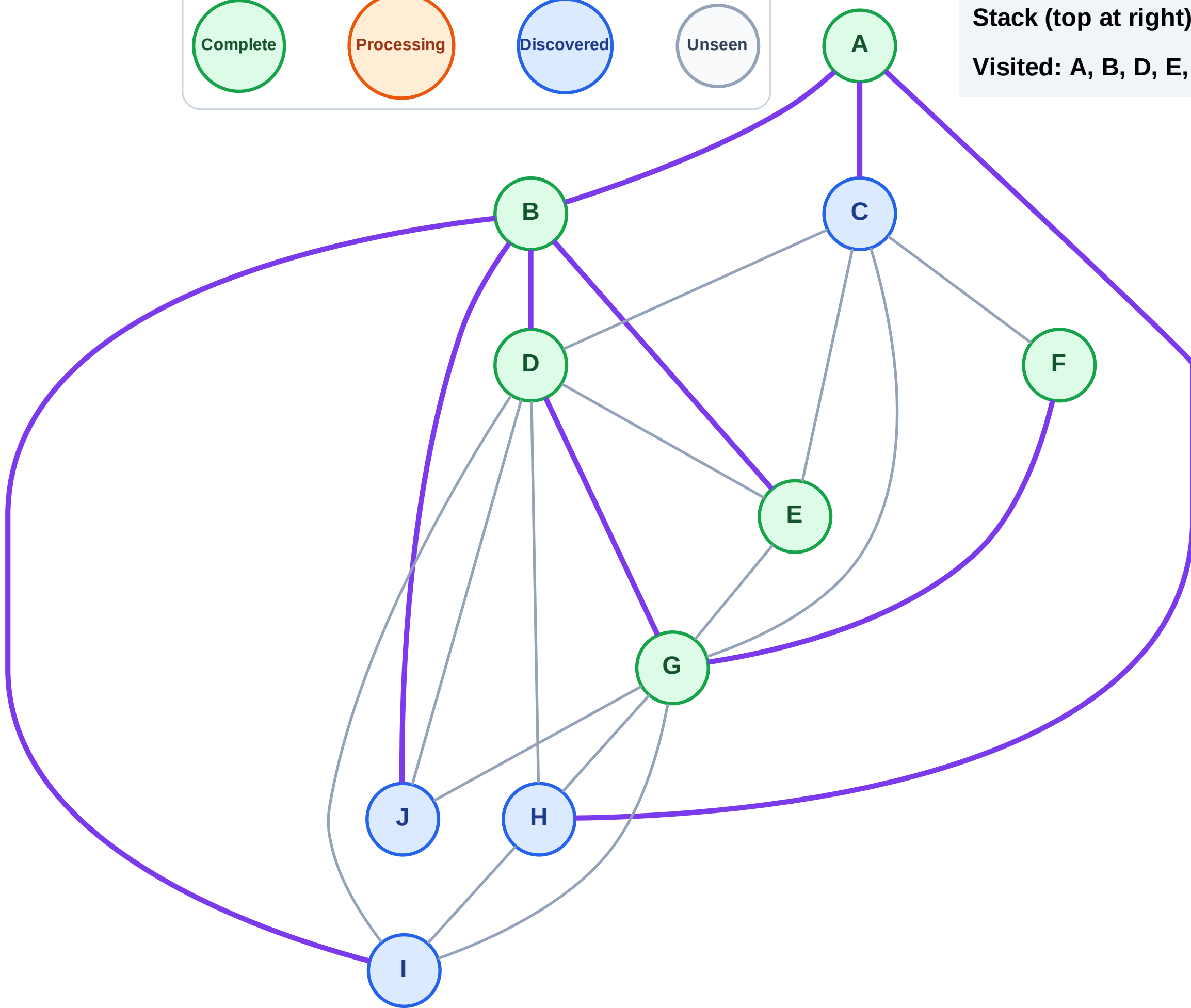
DFS Traversal

Step 13: No new neighbors from E

Legend



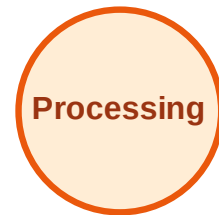
Stack (top at right): H | C | J | I
Visited: A, B, D, E, F, G



DFS Traversal

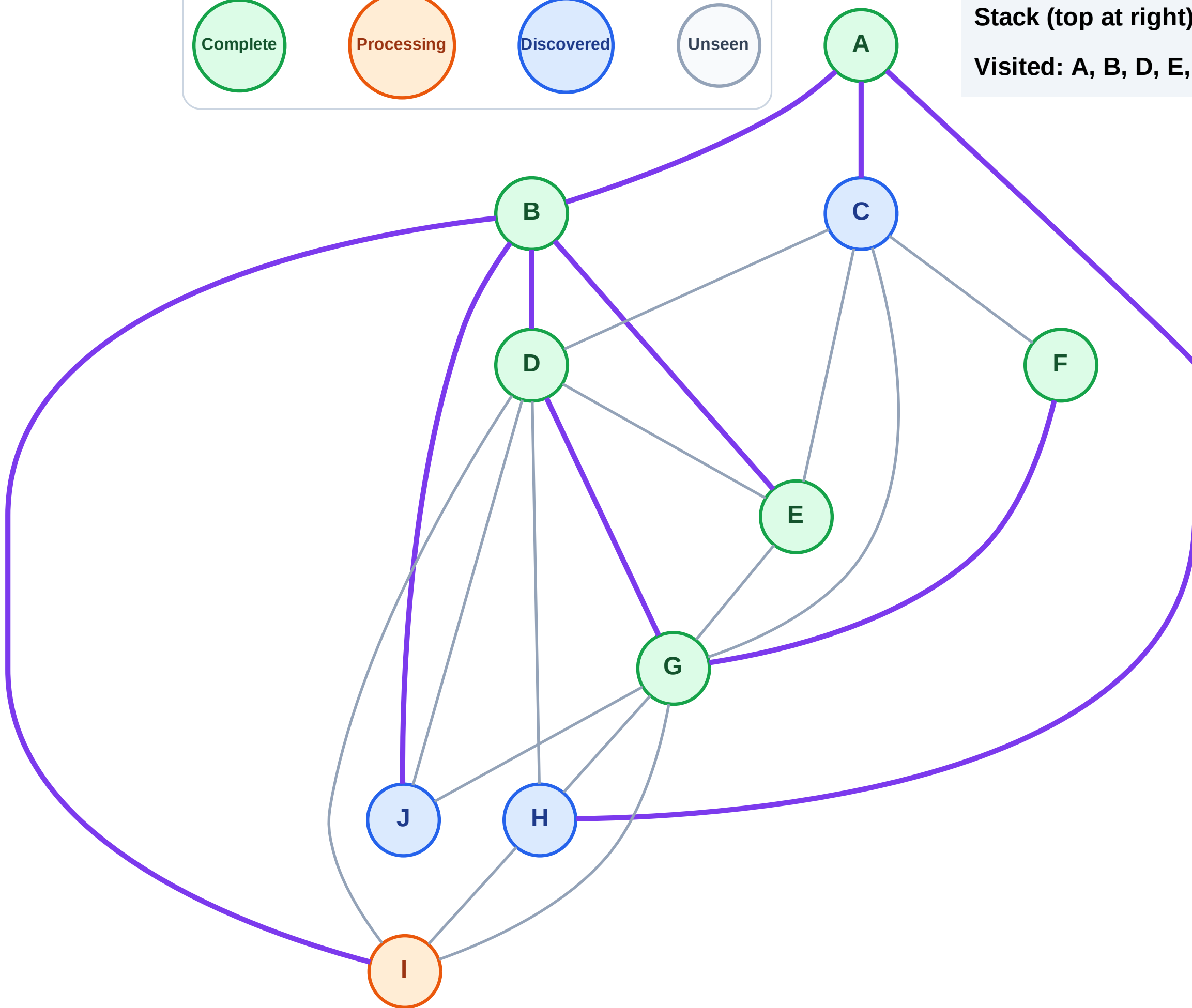
Step 14: Process node I

Legend



Stack (top at right): H | C | J

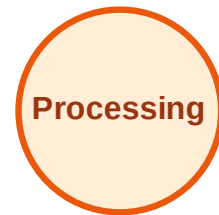
Visited: A, B, D, E, F, G



DFS Traversal

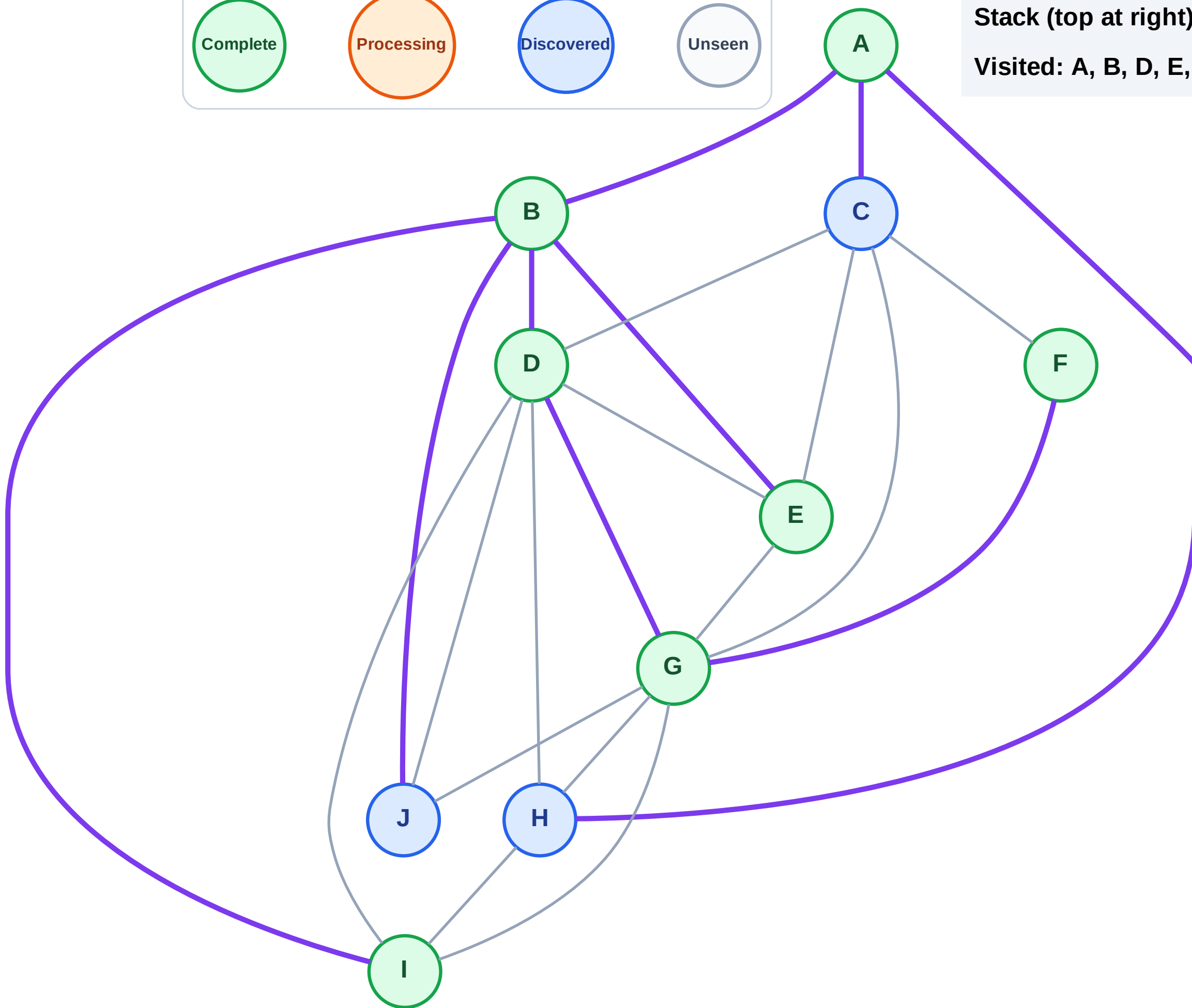
Step 15: No new neighbors from I

Legend



Stack (top at right): H | C | J

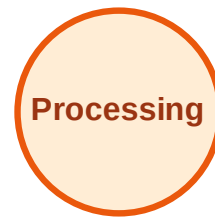
Visited: A, B, D, E, F, G, I



DFS Traversal

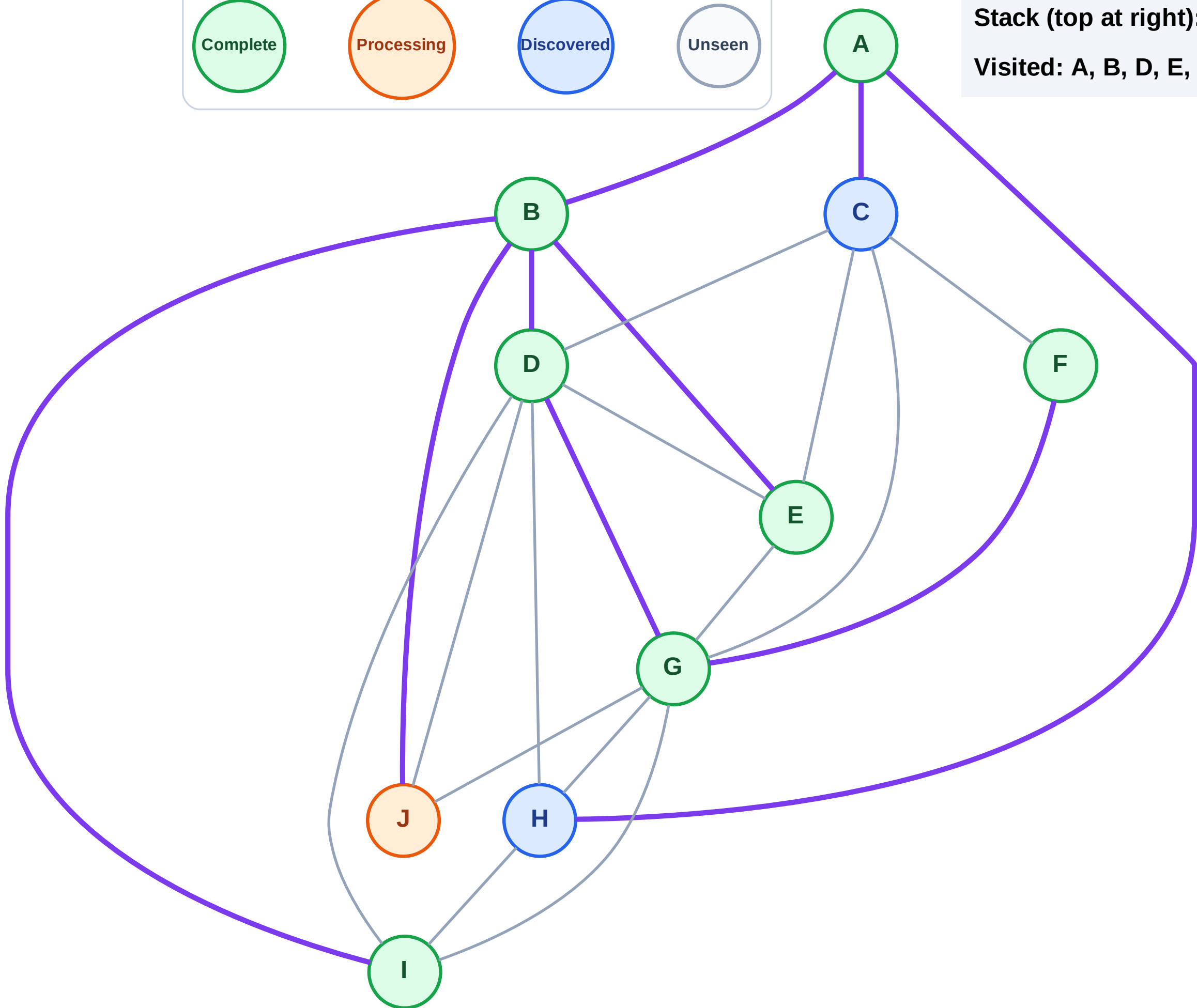
Step 16: Process node J

Legend



Stack (top at right): H | C

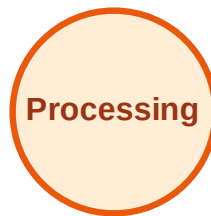
Visited: A, B, D, E, F, G, I



DFS Traversal

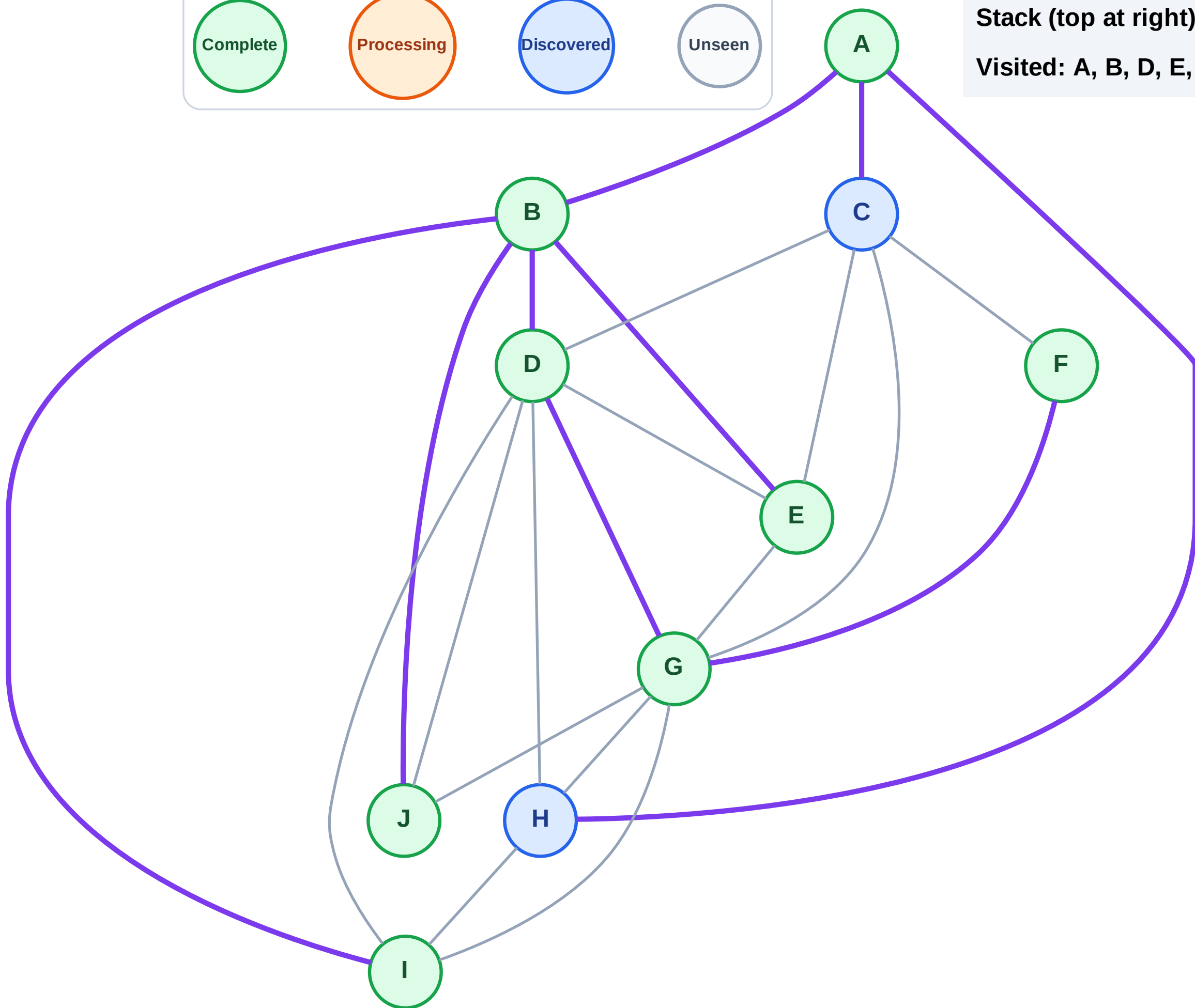
Step 17: No new neighbors from J

Legend



Stack (top at right): H | C

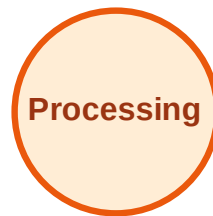
Visited: A, B, D, E, F, G, I, J



DFS Traversal

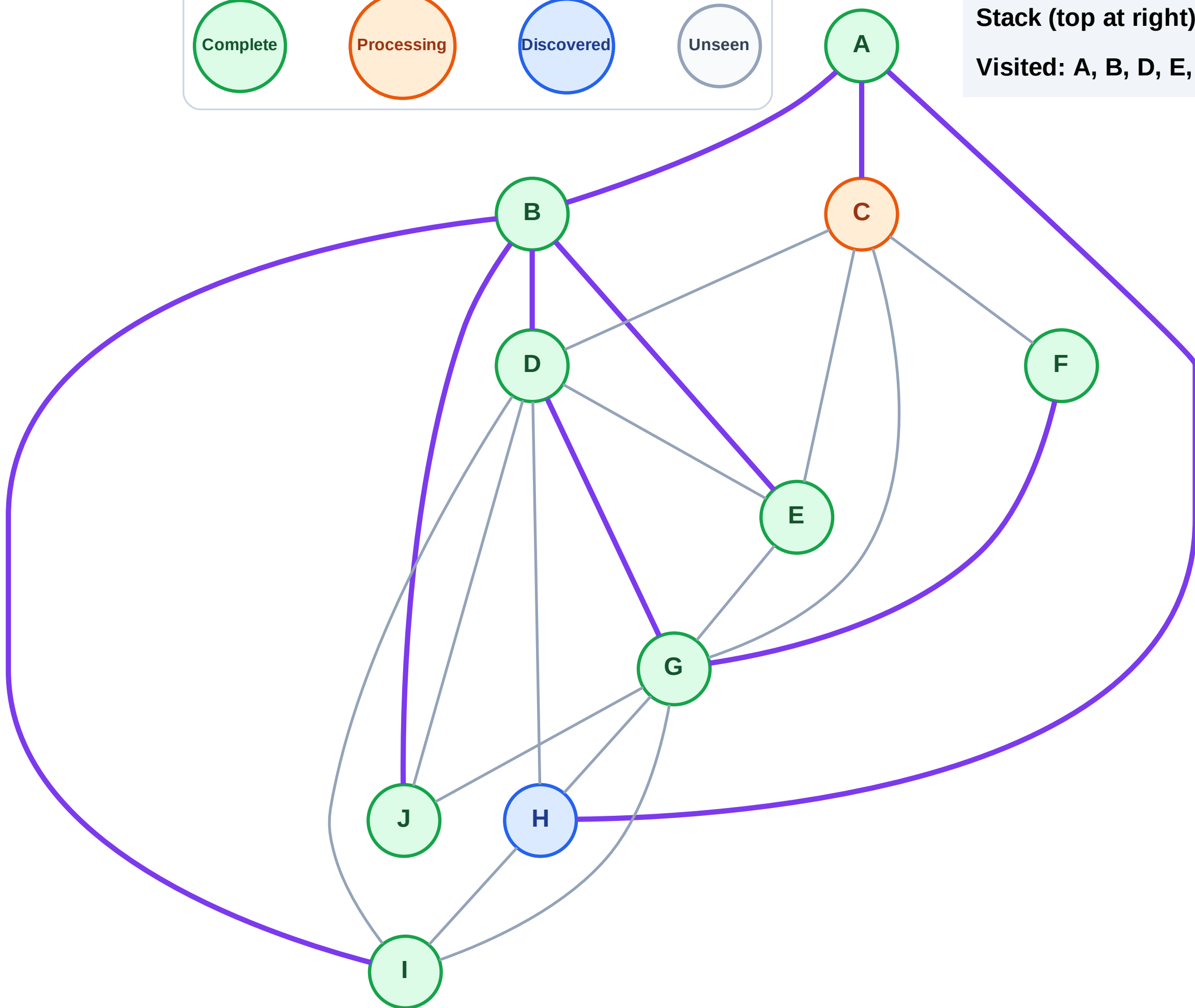
Step 18: Process node C

Legend



Stack (top at right): H

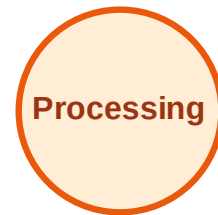
Visited: A, B, D, E, F, G, I, J



DFS Traversal

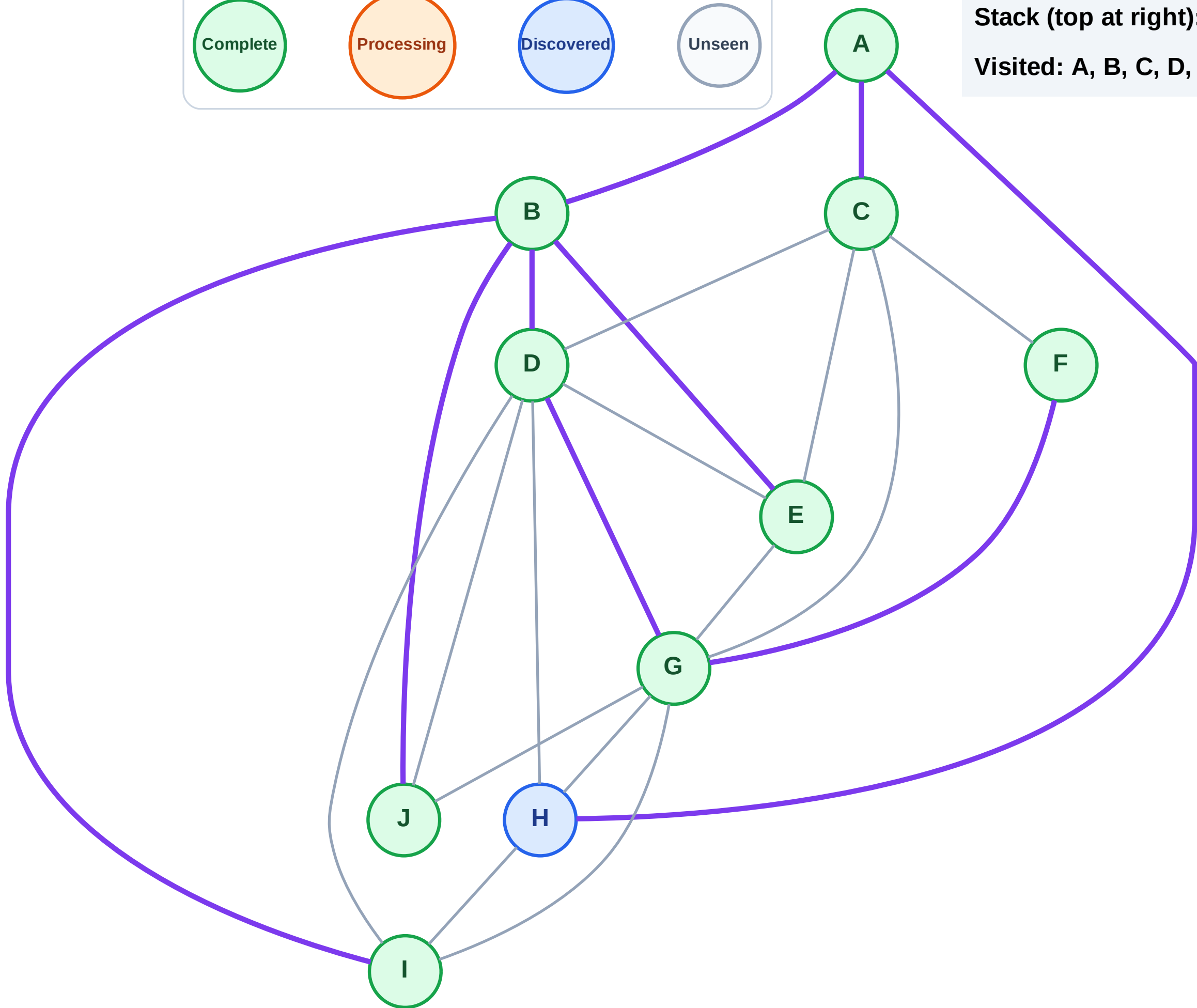
Step 19: No new neighbors from C

Legend



Stack (top at right): H

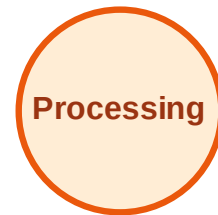
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

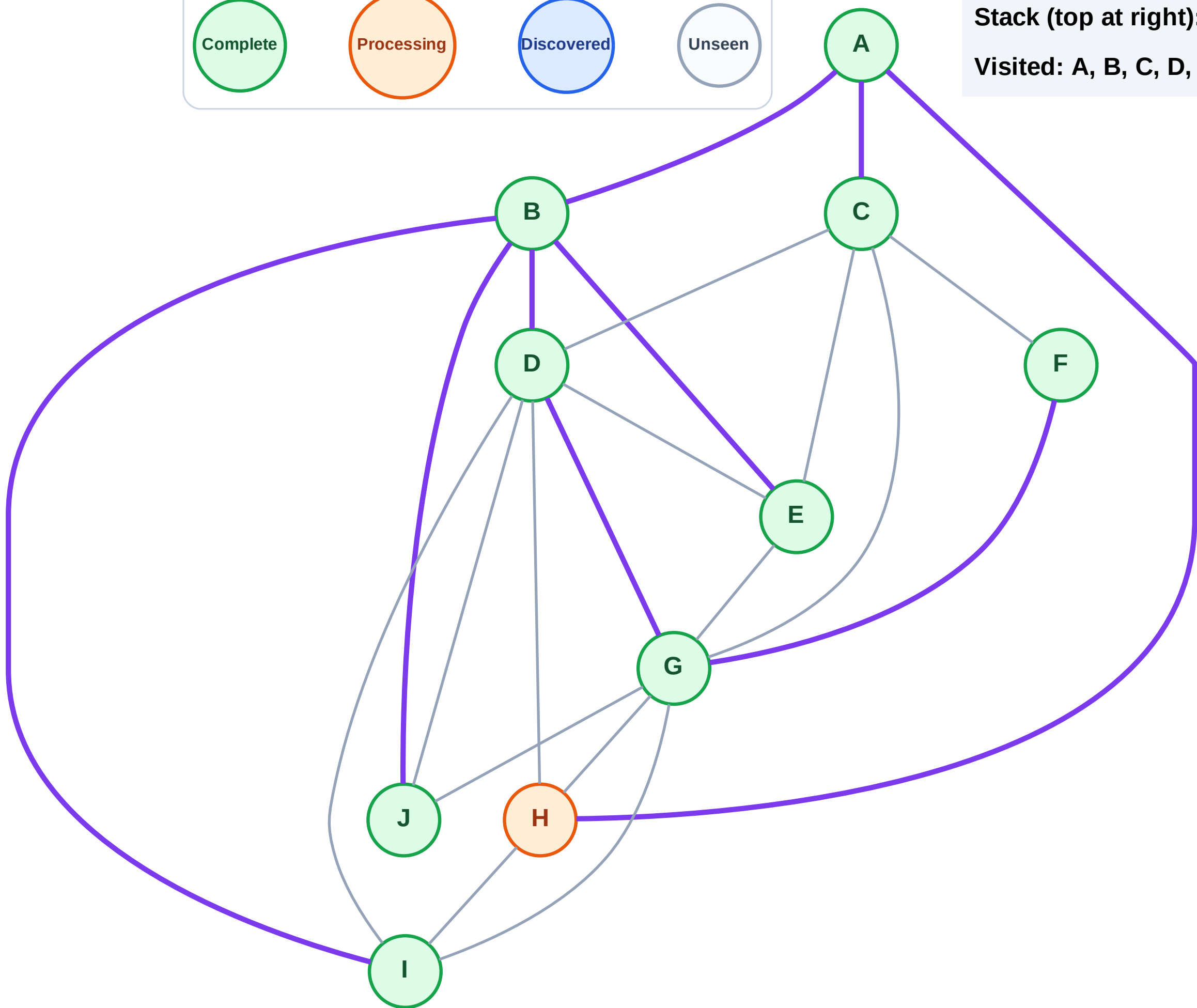
Step 20: Process node H

Legend



Stack (top at right): (empty)

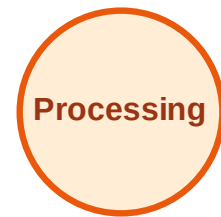
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

